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Module 8:

Schaltungen variabler Frequenz

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1 Repetition

In order to describe the frequency- and time-dependent behaviour of electrical networks, this chapter briefly summarises some important fundamentals.

To work through this chapter, you will need knowledge of basic electrical quantities, linear passive components R, L and C, network calculations, and complex alternating current calculations.

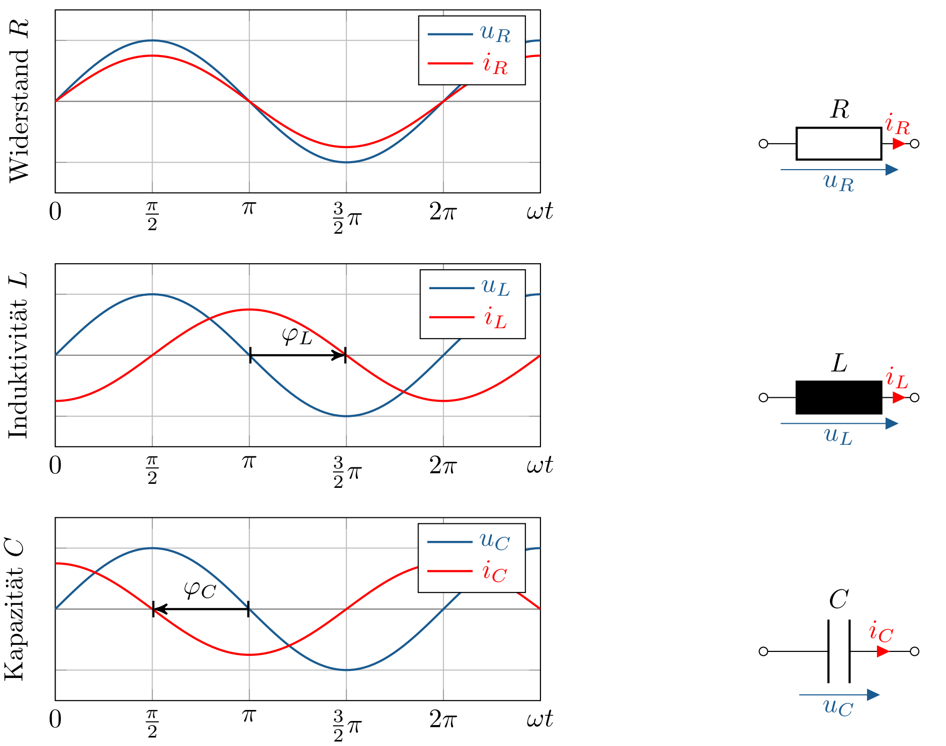
1.1 Frequency dependence of electrical components

Purely ohmic resistors R cannot store electrical energy. Voltages and currents are proportional to each other and in phase at all times.

This behaviour is independent of time and frequency. Inductances L and capacitances C, on the other hand, can store and release energy. This process is inert (sluggish), meaning it takes a certain amount of time.

As a result, their behaviour is frequency-dependent. Currents and voltages in inductors and capacitors are in a (temporal) differential linear relationship to each other. This results in a phase shift between voltage and current of 90 ° for both types of components.

Figure 1.1 shows the voltage and current curves for R, L and C when excited with an alternating voltage $u_q = U \cdot \sin(\omega t)$ for comparison.



(a) Current and voltage curves

(b) Circuit symbols

Figure 1.1: Phase shift at R, L and C

Due to the phase shift in L and C, their power (energy absorption and release) oscillates, but averages out to zero over a period. Inductances and capacitances cannot therefore perform active power, only reactive power, which is why they are also called reactive resistances.

1.1.1 Behaviour of inductances and capacitances

Inductors, as ideal components, store energy in the magnetic field through the effect of (self-)induction. Capacitors, as ideal components, store energy in the electric field. Both effects are described by the law of induction and Gauss's law, respectively.

Table 1.1 lists the most important differences in the behaviour of inductors and capacitors in qualitative terms as an overview.

Table 1.1: Comparison of inductance and capacitance in behaviour

	Inductance	**Capacity**
Law	Induction law	Gauss's law
Energy storage	in the magnetic field	in the electric field
continuous	current	voltage
with DC voltage	short circuit	open
with high frequency	open	short circuit

The (self-)inductance L as a property can be simplified as „inertia "of the current. Currents in inductors are continuous and lag behind the voltage. At high frequencies, the inductance blocks; at DC voltage, it behaves like a short circuit.

In contrast, capacitance C can be simplified as „inertia "of the voltage. Voltages in capacitors are constant and lag behind the current. With direct current, capacitance blocks the current; at high frequencies, it behaves like a short circuit.

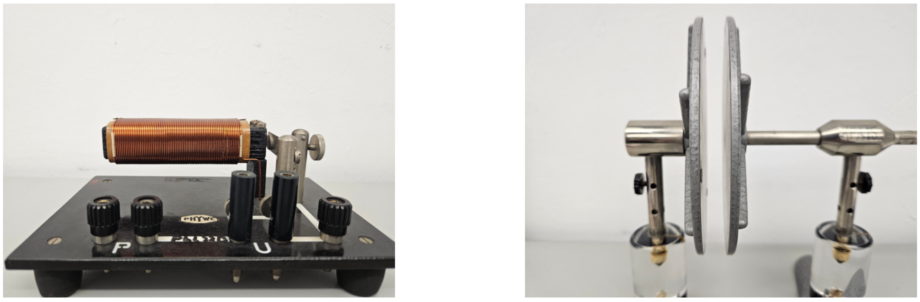


Figure 1.2: Real inductance (coil, left) and capacitance (capacitor, right)

Figure 1.2 shows real components that implement inductance and capacitance. Shown are a coil (inductance) and a capacitor (capacitance).

1.1.2 Comparison of linear two-pole elements R, L and C

1.2 shows a comparison of the quantities R, L and C and summarises their most important properties.

1.2 Double-pole (four-pole)

We already examined the frequency dependence of single ports (dipoles) in Module 3. Using complex AC calculations, we were able to determine the frequency-dependent AC resistance (impedance) $\underline{Z}$ of individual single ports. By applying the complex current and voltage divider rules, we were also

Table 1.2: Comparison of linear components R, L, C

Size	**Generally**	**El. resistance**	**Inductance**	**Capacity**
Symbol	Zweipol Eintor	R	L	C
Unit	$[Form.s.] = \mathrm{Unit}$	$[R] = \Omega$ (Ohm)	$[L] = \mathrm{H}$ (Henry)	$[C] = \mathrm{F}$ (Farad)
Time domain	$\frac{\mathrm{d}}{\mathrm{d}t}$ or $\int \mathrm{d}t$	$u_\mathrm{R} = R \cdot i_\mathrm{R}$	$u_\mathrm{L} = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i_\mathrm{L}$	$i_\mathrm{C} = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_\mathrm{C}$
frequency range	$\mathrm{j}\omega$ or $\frac{1}{\mathrm{j}\omega}$	$\underline{U}_\mathrm{R} = R \cdot \underline{I}_\mathrm{R}$	$\underline{U}_\mathrm{L} = \mathrm{j}\omega L \cdot \underline{I}_\mathrm{L}$	$\underline{I}_\mathrm{C} = \mathrm{j}\omega C \cdot \underline{U}_\mathrm{C}$
Impedance	$\underline{Z} = \frac{\underline{U}}{\underline{I}}$	$\underline{Z}_\mathrm{R} = R$	$\underline{Z}_\mathrm{L} = \mathrm{j}\omega L$	$\underline{Z}_\mathrm{C} = -\mathrm{j}\frac{1}{\omega C}$
Active part	$R = \mathrm{Re}\{\underline{Z}\}$	$R = \frac{U}{I}$	0	0
Blind part	$X = \mathrm{Im}\{\underline{Z}\}$	0	$X_\mathrm{L} = \omega L$	$X_\mathrm{C} = -\frac{1}{\omega C}$

able to determine the total impedance of linear two-terminal networks. This applies to networks that consist only of linear components such as R, L and C.

Figure 1.3: Comparison: Circuit symbols for a general two-pole and four-pole circuit

Two-port (four-terminal) networks are an extension of single-port (two-terminal) networks. They have an input and an output side with respective input and output variables. The two-port model is well suited for describing (frequency-dependent) transmission characteristics of electrical networks.

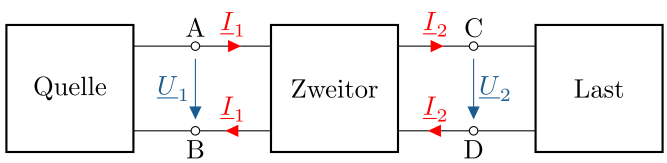


Figure 1.4: Connection of a load to the source via a two-port network.

In the simplest case, as shown in 1.4, a source (single-port) is connected to a load (single-port) via a two-port. Typical two-ports are, for example, amplifiers, filters or transformers.

2 Frequency response, amplitude response, phase response

In this chapter, we will deal with the frequency response, amplitude response and phase response of two-port networks. These are used to describe the frequency-dependent behaviour of electrical networks in the form of two-port networks, as shown in Figure 2.1. Using the example of simple passive filter circuits, we will introduce the terms frequency response, amplitude response and phase response and explain their meaning and application.

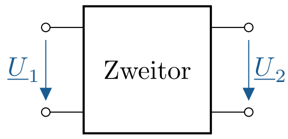


Figure 2.1: Schaltsymbol eines Zweitors (Vierpols) mit Eingang (links) und Ausgang (rechts)

Learning objectives: Frequency response, amplitude response, phase response

Students learn:

- Determine frequency responses algebraically based on circuit topology
- Understand how simple high-pass and low-pass filters work
- Analyse the limiting behaviour of four-poles based on their frequency responses

2.1 Definition of frequency response, amplitude response and phase response

The **frequency response** $\underline{F}(\mathrm{j}\omega)$, also known as the complex amplitude response, describes the ratio of the output signal to the input signal of a linear time-invariant system (LTI system) under sinusoidal excitation. It offers a way to investigate the frequency-dependent behaviour of two-port networks, is a special case of the Laplace transfer function from system theory, and is defined as follows:

$$\underline{F}(\mathrm{j}\omega) = \frac{\underline{U}_2}{\underline{U}_1} \tag{2.1}$$

Due to the linearity of LZI systems, the frequency and sine waveform are retained at the output. The amplitude and phase may change from the input to the output depending on the frequency. This is particularly evident in the notation in polar coordinates.

$$\underline{F}(\mathrm{j}\omega) = A(\omega) \cdot \mathrm{e}^{\mathrm{j}\varphi(\omega)} \tag{2.2}$$

The **amplitude response** $A(\omega)$ describes the relative amplitude change between output and input voltage and corresponds to the magnitude of the frequency response. The **phase response** $\varphi(\omega)$, on the other hand, describes the absolute phase change between output and input voltage and corresponds to the phase of the frequency response. Using $\underline{F}(\mathrm{j}\omega)$, we can define both as follows:

$$A(\omega) = |\underline{F}(\mathrm{j}\omega)| \qquad = \sqrt{(\mathrm{Re}\{\underline{F}(\mathrm{j}\omega)\})^2 + (\mathrm{Im}\{\underline{F}(\mathrm{j}\omega)\})^2} \qquad = \frac{|\underline{U}_2|}{|\underline{U}_1|} \tag{2.3}$$

$$\varphi(\omega) = \angle\underline{F}(\mathrm{j}\omega) \qquad = \arctan\left(\frac{\mathrm{Im}\{\underline{F}(\mathrm{j}\omega)\}}{\mathrm{Re}\{\underline{F}(\mathrm{j}\omega)\}}\right) \qquad = \angle\underline{U}_2 - \angle\underline{U}_1 \tag{2.4}$$

2.2 Frequency response using the example of a simple low-pass filter

The two-port network in Figure 2.2 represents a simple low-pass filter (low-pass for short). The name low-pass is derived from the behaviour that low-frequency signals pass through almost unchanged, while high-frequency signals are strongly attenuated (filtered). The functionality and behaviour of the low-pass filter can be clearly illustrated by means of a frequency response analysis.

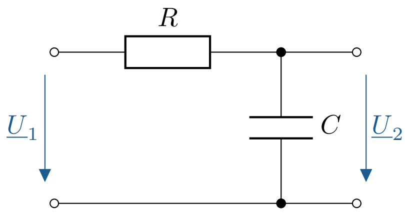


Abbildung 2.2: Schaltbild, RC-Tiefpass 1. Ordnung

The low-pass filter shown consists of a series connection of R and C with input voltage $\underline{U}_1$ across both elements and output voltage $\underline{U}_2$ across C. Since the individual elements R and C are linear, time-invariant (LTI) components, the low-pass filter composed of them is also an LTI system. This means that when the circuit is excited with a sinusoidal input voltage $u_1(t)$ (left), the output voltage $u_2(t)$ (right) is also sinusoidal and has the same frequency as the input voltage.

$$\begin{aligned} & u_1(t) = \hat{U}_1 \cdot \sin(\omega t) \\ \xrightarrow{LZI}\ & u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi) \end{aligned} \tag{2.5}$$

The LZI property is a prerequisite for determining the frequency response and applying complex AC calculations. An additional prerequisite is the assumption of sinusoidal excitation in the steady state, which we use as a basis for our analysis. Instead of the input and output voltages $u_1(t)$ and $u_2(t)$ in the time domain, we consider the corresponding complex input and output voltages $\underline{U}_1$ and $\underline{U}_2$ in the frequency domain, as shown in the circuit diagram.

According to equation 2.1, the frequency response $\underline{F}(\mathrm{j}\omega)$ corresponds to the complex voltage ratio of output to input voltage $\underline{U}_2/\underline{U}_1$. According to the complex voltage divider rule, the voltage ratio corresponds to the ratio of the impedance at the output (capacitance impedance) and the impedance at the input (sum of the capacitance and resistance impedances), or $\frac{\underline{Z}_C}{R+\underline{Z}_C}$ for short.

Equation 2.6 shows the determined frequency response of the low-pass filter. For simplification, the denominator and numerator have been rationalised, the denominator sorted and the term (optionally) transformed into Cartesian coordinates. (Compare annotated side calculation 6.1):

$$\begin{aligned} \underline{F}(\mathrm{j}\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}} = \frac{1}{\mathrm{j}\omega CR + 1} \\ = \frac{1}{1 + \mathrm{j}\omega CR} = \frac{1 - \mathrm{j}\omega CR}{1 + (\omega CR)^2} \end{aligned} \tag{2.6}$$

We have thus determined $\underline{F}(\mathrm{j}\omega)$ as a function of ω and the component values C and R. Using absolute values, we can determine the amplitude response according to (2.3) and the phase response by calculating the phase according to (2.4).

The form $\frac{1}{1+\mathrm{j}\omega CR}$ is suitable for calculating the amount.

Since the magnitude in the numerator is directly readable as 1, only the magnitude in the denominator needs to be determined, as shown in (2.7). Alternatively, the Cartesian form from (2.6) can also be used in (2.3).

For the phase shift $\varphi(\omega)$, the Cartesian notation in the form $\frac{1-\mathrm{j}\omega CR}{1+(\omega CR)^2}$ is suitable. In this notation, the real and imaginary parts can be easily read. As a purely real quantity, the denominator has no influence on the phase as shown in (2.8).

$$A(\omega) = |\underline{F}(\mathrm{j}\omega)| = \left|\frac{1}{1+\mathrm{j}\omega CR}\right| = \frac{|1|}{|1+\mathrm{j}\omega CR|} = \frac{1}{\sqrt{1+(\omega CR)^2}} \tag{2.7}$$

$$\varphi(\omega) = \angle\underline{F}(\mathrm{j}\omega) = \angle\left(\frac{1-\mathrm{j}\omega CR}{\cancel{1+(\omega CR)^2}}\right) = \arctan\left(\frac{\frac{-\omega CR}{\cancel{1+(\omega CR)^2}}}{\frac{1}{\cancel{1+(\omega CR)^2}}}\right) = \arctan\left(-\omega CR\right) \tag{2.8}$$

Alternatively, the phase response can also be determined from the non-Cartesian form of the frequency response. The phase response is calculated from the phase angle of the numerator subtracted from the phase angle of the denominator:

$$\varphi(\omega) = \angle\left(\frac{1}{1+\mathrm{j}\omega CR}\right) = \underbrace{\arctan\left(\frac{0}{\cancel{1}}\right)}_{\varphi_{Zaehler}} - \underbrace{\arctan\left(\frac{\omega CR}{\cancel{1}}\right)}_{\varphi_{Nenner}} = \arctan\left(-\omega CR\right) \tag{2.9}$$

This approach is particularly useful if the numerator and/or phase can be read directly from the counter. This saves one calculation step when converting to Cartesian coordinates.

The following digression explains in more detail how to consider the numerator and denominator of frequency responses separately:

Digression: Calculating complex numbers in (Cartesian) Cartesian and polar coordinates

The frequency response usually takes the form of a complex fractional rational fraction. This means that both the denominator and the numerator can be represented as complex polynomials:

$$\underline{F}(\mathrm{j}\omega) = \frac{\underline{F}_Z(\mathrm{j}\omega)}{\underline{F}_N(\mathrm{j}\omega)} = \frac{a_n\omega^n + \cdots + a_1\omega + a_0}{b_m\omega^m + \cdots + b_1\omega + b_0} \qquad \text{mit} \qquad a_i, b_j \in \mathbb{C} \qquad n,m,\omega \in \mathbb{R} \tag{2.10}$$

If $\underline{F}(\mathrm{j}\omega)$ is now available as a complex fraction with numerator and denominator in Cartesian coordinates, we can determine $A(\omega)$ and $\varphi(\omega)$ by separately calculating the magnitude and phase of the numerator and denominator. For this purpose, the formulas from (2.3) and (2.4) are applied for the numerator (index Z) and denominator (index N). Since the amplitude and phase response, as shown in (2.2), represent the polar coordinates (amplitude and phase) of the frequency response, the following calculation scheme results:

$$\underline{F}(\mathrm{j}\omega) = \frac{\underline{F}_Z(\mathrm{j}\omega)}{\underline{F}_N(\mathrm{j}\omega)} = \frac{|\underline{F}_Z| \cdot \mathrm{e}^{\mathrm{j}\angle \underline{F}_Z}}{|\underline{F}_N| \cdot \mathrm{e}^{\mathrm{j}\angle \underline{F}_N}} = \frac{A_Z}{A_N} \cdot \mathrm{e}^{\mathrm{j}(\varphi_Z - \varphi_N)} \tag{2.11}$$

$$A(\omega) = |\underline{F}(\mathrm{j}\omega)| = \frac{|\underline{F}_Z|}{|\underline{F}_N|} = \frac{A_Z}{A_N} \tag{2.12}$$

$$\varphi(\omega) = \angle \underline{F}(\mathrm{j}\omega) = \angle \underline{F}_Z - \angle \underline{F}_N = \varphi_Z - \varphi_N \tag{2.13}$$

In general, polar coordinates (with magnitude and argument) are useful for multiplying and dividing complex numbers. Multiplication/division of complex numbers leads to multiplication/division of the magnitudes and addition/subtraction of the arguments. For addition and subtraction, the Cartesian form with real and imaginary parts is better suited for calculations, as these can be added/subtracted directly.

2.3 Boundary behaviour using the example of a simple low-pass filter

This chapter examines the boundary behaviour of the first-order low-pass filter from the previous chapter (see Figure 2.2) for very high and very low frequencies. To this end, the boundary values of the amplitude response $A(\omega)$ and the phase response $\varphi(\omega)$ are determined, and the results are described, interpreted and presented graphically. **Limit values:**

For the amplitude response from equation 2.7, the following limit values result for $f \to 0$ and $f \to \infty$:

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\lim_{\omega \to 0} A(\omega) = \frac{1}{1} = 1 \tag{2.14}$$

$$\lim_{\omega \to \infty} A(\omega) = \frac{1}{\sqrt{\infty}} = 0$$

The phase shift from equation 2.8 yields the following limit values:

$$\varphi(\omega) = \arctan(-\omega CR)$$

$$\lim_{\omega \to 0} \varphi(\omega) = \arctan(0) = 0^\circ \tag{2.15}$$

$$\lim_{\omega \to \infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$$

Equation 2.14 shows that the low-pass filter hardly attenuates at very low frequencies ($A(\omega) \to 1$), but attenuates strongly at very high frequencies ($A(\omega) \to 0$).

Equation 2.15 shows that the low-pass filter exhibits hardly any phase shift at very low frequencies (with hardly any attenuation) ($\varphi(\omega) \to 0^\circ$), but exhibits a phase shift of up to -90° at very high frequencies (strong attenuation).

Explanation of the limit behaviour based on the circuit diagram:

The boundary behaviour with regard to amplitude change and phase shift can be explained well using the circuit diagram 2.2.

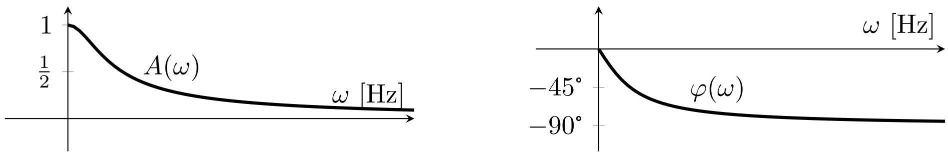
In the limiting case of a DC voltage ($f = 0$) at the input, the capacitance ($X_C \to \infty$) blocks the current. Its behaviour here corresponds to two open terminals, which means that the entire input voltage is applied to the output U_2 across the capacitance C. The relative amplitude change is therefore 1 (undamped) and the phase shift 0° (in phase).

At very high frequencies ($f \to \infty$), the capacitance behaves like a short circuit ($X_C \to 0$). This causes the output voltage to approach zero (strong attenuation, $A \to 0$) and the input voltage to be almost entirely across R. The input voltage and current are therefore approximately in phase. The phase shift of the output voltage across the capacitance to the input voltage is therefore approximately -90°.

Visualisation:

To get an idea of the behaviour of amplitude and phase response, we can plot the functions graphically.

The amplitude response $A(\omega)$ and the phase response $\varphi(\omega)$ are shown in Figures 2.3a and 2.3b. Both graphs are shown with linear scales, whereby the scaling of both x-axes for ω [Hz] are identical. A change in the factor CR in a linear representation merely leads to a compression or stretching of the graphs in the x-direction, which is why the values were not explicitly specified.



(a) Amplitude response $A(\omega)$, linear axes (b) Phase shift $\varphi(\omega)$, linear axes

Figure 2.3: Linear representation of the amplitude response and phase response of a first-order low-pass filter

Chapter 3 introduces the representation of frequency response using a logarithmic scale. The definition of the **cut-off frequency** f_g (or the cut-off circular frequency ω_g) for dividing the frequency range into a **passband** and a **stopband** is also explained there.

Digression on the derivation of the graph shape:

The graph shape of the amplitude response $A(\omega)$ in linear representation can be explained by the fact that it approaches the horizontal straight line $A = 1$ for very low frequencies ($f \to 0$) and approaches the power function $\frac{1}{\omega CR}$ for very high frequencies ($f \to \infty$). Figure 2.4 shows these approximations for $A(x) = \frac{1}{\sqrt{1+x^2}}$ with $x = \omega CR$.

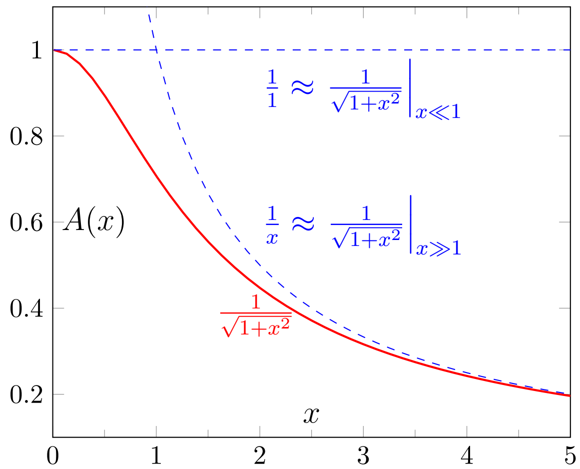


Figure 2.4: Amplitude response of a first-order low-pass filter, linear axes, approximations with $x = \omega CR$

The shape of the phase response can be easily derived from the shape of a (arc) tangent function. Figure 2.5 shows the tangent function and its inverse function, the arc tangent, as well as their respective reflections on the y-axis in a linear representation.

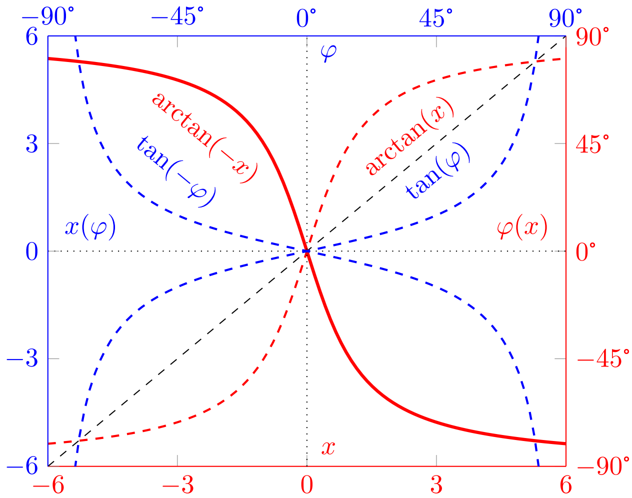


Figure 2.5: Phase response of a first-order low-pass filter, linear axes, comparison of tangent/arc tangent

The inverse of the respective function can be obtained by reflecting the graph about the bisector (black dotted line) and swapping the x- and y-axes, provided that the respective function is continuously monotonically increasing or continuously monotonically decreasing. The tangent is therefore limited to the angle range −90° to +90° for the formation of the arc tangent with a value range from $-\infty$ to $+\infty$. Due to the non-representable pole points, the value range shown is limited to −6 to +6.

The phase response of the first-order low-pass filter corresponds to the plotted $\arctan(-x)$ for $x = \omega CR > 0$. Since there are no negative frequencies in physics, $\omega > 0 \rightarrow x > 0$ applies, which means that the value range of the phase response only covers half the value range of the arctangent, i.e. from −90° to 0°.

2.4 Frequency response using the example of a simple high-pass filter

A high-pass filter (high pass for short) works similarly to a low-pass filter. However, as the name suggests, high-pass filters allow high-frequency signals to pass through almost unattenuated (unfiltered), while low-frequency signals are strongly attenuated (filtered).

Figure 2.6 shows an example of a first-order RC high-pass filter, which will be examined in more detail below. The high-pass filter consists of a simple series connection of resistor R and capacitor C. The input voltage $\underline{U}_1$ is applied across R and C in series, while the output voltage $\underline{U}_2$ is tapped across R. The setup is identical to the first-order RC low-pass filter, except that the output voltage is applied across R instead of C.

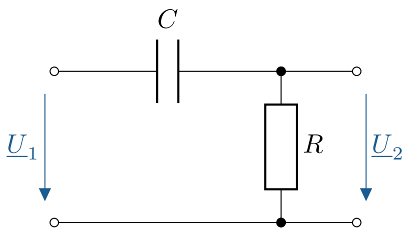


Figure 2.6: Circuit diagram, first-order RC high-pass filter

The frequency response is derived in the same way as for the low-pass filter in section 2.2. Since the high-pass filter is an LZI system, the following applies:

$$\begin{aligned} & u_1(t) = \hat{U}_1 \cdot \sin(\omega t) \\ \xrightarrow{LZI} \; & u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi) \end{aligned} \tag{2.16}$$

This means that for a sinusoidal input voltage $u_1(t)$, the output voltage $u_2(t)$ is also sinusoidal with the same frequency as the input voltage.

The frequency response can thus be determined according to equation 2.1 by setting the complex alternating voltages $\underline{U}_2$ at the output and $\underline{U}_1$ at the input in relation to each other. This gives us:

$$\begin{aligned} \underline{F}(\mathrm{j}\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{R}{R + \frac{1}{\mathrm{j}\omega C}} = \frac{1}{1 + \frac{1}{\mathrm{j}\omega CR}} \\ &= \frac{1}{1 - \mathrm{j}\,\frac{1}{\omega CR}} = \frac{1 + \mathrm{j}\,\frac{1}{\omega CR}}{1 + \frac{1}{(\omega CR)^2}} \end{aligned} \tag{2.17}$$

The following equations result for the amplitude response $A(\omega)$ and the phase response $\varphi(\omega)$:

$$A(\omega) = |\underline{F}(\mathrm{j}\omega)| = \left| \frac{1}{1 - \mathrm{j}\frac{1}{\omega CR}} \right| = \frac{1}{\sqrt{1 + \frac{1}{(\omega CR)^2}}} \tag{2.18}$$

$$\varphi(\omega) = \angle \underline{F}(\mathrm{j}\omega) = \angle \left(\frac{1 + \mathrm{j}\frac{1}{\omega CR}}{1 + \frac{1}{(\omega CR)^2}} \right) = \arctan\left(\frac{1}{\omega CR} \right) \tag{2.19}$$

2.5 Boundary behaviour using the example of a simple high-pass filter

The boundary behaviour of the first-order high-pass filter from Chapter 2.4 is examined in this chapter for very high and very low frequencies. The limit values of the amplitude response $A(\omega)$ and the phase response $\varphi(\omega)$ are determined in the same way as for the low-pass filter described in Chapter 2.3.

Limit value:

The amplitude response from equation 2.18 yields the following limit values for $f \to 0$ and $f \to \infty$:

$$\begin{aligned} A(\omega) &= \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}} \\ \lim_{\omega \to \infty} A(\omega) &= \frac{1}{\infty} &&= 0 \\ \lim_{\omega \to 0} A(\omega) &= \frac{1}{\sqrt{1}} &&= 1 \end{aligned} \tag{2.20}$$

For the phase response from equation 2.19, the following applies analogously:

$$\begin{aligned} \varphi(\omega) &= \arctan\left(\frac{1}{\omega CR} \right) \\ \lim_{\omega \to 0} \varphi(\omega) &= \arctan(\infty) &&= +90^\circ \\ \lim_{\omega \to \infty} \varphi(\omega) &= \arctan(0) &&= 0^\circ \end{aligned} \tag{2.21}$$

This means that at very low frequencies, the high-pass filter attenuates strongly ($A(\omega) \to 0$) with a phase shift of up to +90° and at very high frequencies, it attenuates hardly at all ($A(\omega) \to 1$) with a phase shift of up to 0°.

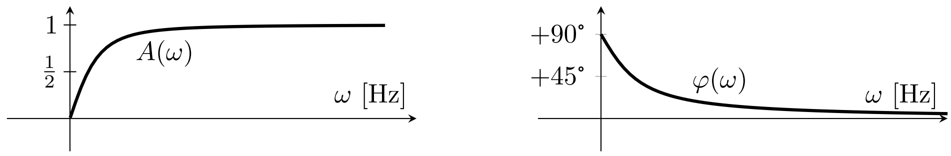
This boundary behaviour can be explained well using the circuit diagram 2.6. At DC voltage ($f = 0$), the capacitance blocks, causing the entire input voltage to drop across it.

At very low frequencies ($f \to 0$), the input voltage is approximately completely across the capacitance. The current leads the input voltage by up to +90°, resulting in a positive phase shift up to +90° to the output voltage across the resistor.

At very high frequencies ($f \to \infty$), the capacitance behaves like a short circuit, causing the input voltage to be approximately completely across the resistor at the output. As a result, the output and input voltages are approximately equal in amplitude and phase.

Visualisation:

Figure 2.7 shows the amplitude response $A(\omega)$ and the phase response $\varphi(\omega)$ of the first-order highpass filter in linear representation with the same scaling and the same value range for the x-axes.



(a) Amplitude response $A(\omega)$, linear axes (b) Phase response $\varphi(\omega)$, linear axes

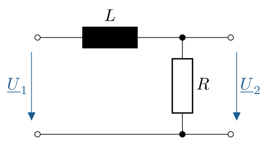
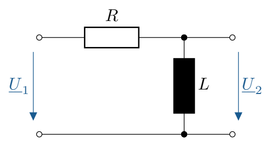
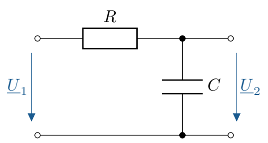
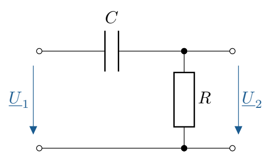
Figure 2.7: Linear representation of the amplitude response and phase response of a first-order high-pass filter

A logarithmic representation of the frequency response is introduced in Chapter 3.4.2.

2.6 Comparison of first-order passive filters with L and R

Low-pass and high-pass filters can also be implemented using an inductance L and a resistor R. Table 2.1 compares the circuit diagrams for first-order RL filters and their RC counterparts. The formula for the frequency response is also listed for each circuit diagram.

Table 2.1: Comparison of circuit diagram and frequency response for first-order RL and RC filters

	Low-pass filter	$\underline{F}(\mathrm{j}\omega)$	**High-pass filter**	$\underline{F}(\mathrm{j}\omega)$	ω_g
RL		$\dfrac{1}{1+\mathrm{j}\omega\frac{L}{R}}$		$\dfrac{1}{1-\mathrm{j}\frac{R}{\omega L}}$	$\dfrac{R}{L}$
RC		$\dfrac{1}{1+\mathrm{j}\omega CR}$		$\dfrac{1}{1-\mathrm{j}\frac{1}{\omega CR}}$	$\dfrac{1}{CR}$

A comparison of the circuit diagrams shows that the RL and RC variants differ only in the arrangement of L and R and R and C, respectively. The frequency responses of the RL and RC variants differ only in the factor $\frac{L}{R}$ and CR before ω. As will be explained in more detail in Chapter 3.3, this factor corresponds to the reciprocal of the cutoff frequency ω_g, which is included in the table for the sake of completeness.

2.7 Beispiel Filterschaltung: Aufgabe

Beispiel 2.1: Filterschaltung: Aufgabe

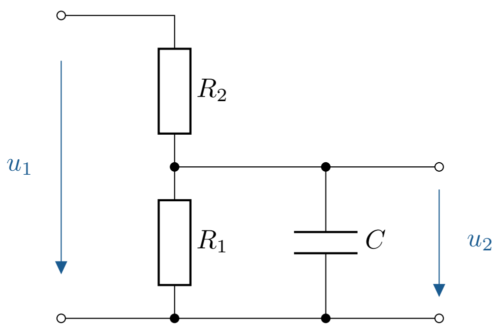


Abbildung 2.8: Filterschaltung, passiv

Aufgabe:
a) Leiten Sie $\underline{F}(\mathrm{j}\omega) = \frac{\underline{U}_2}{\underline{U}_1}$ allgemein her.
b) Zeichnen Sie Betrag und Phase von $\underline{F}(\mathrm{j}\omega)$ für:

$R_1 = 900\ \Omega$

$R_2 = 100\ \Omega$

$C = 1{,}25\ \mu\mathrm{F}$

c) Bei welcher Frequenz f ist $\hat{U}_2 = \frac{1}{2}\hat{U}_1$? Welche Phasenverschiebung besitzt u_2 nun gegenüber u_1?

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t) \qquad \underline{U}_1 = \hat{U}_1$$
$$u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi) \qquad \underline{U}_2 = \hat{U}_2 \cdot \mathrm{e}^{\mathrm{j}\varphi}$$

Beispiel 2.2: Filter circuit: Solution

Solution a) Frequency response in general

$$\underline{F}(\mathrm{j}\omega) = \frac{R_1||\underline{Z}_C}{R_2 + (R_1||\underline{Z}_C)} = \frac{\frac{\frac{R_1}{\mathrm{j}\omega C}}{R_1+\frac{1}{\mathrm{j}\omega C}}}{R_2 + \frac{\frac{R_1}{\mathrm{j}\omega C}}{R_1+\frac{1}{\mathrm{j}\omega C}}} \quad \Bigg| \cdot \frac{R_1 + \frac{1}{\mathrm{j}\omega C}}{R_1 + \frac{1}{\mathrm{j}\omega C}}$$

$$= \frac{\frac{R_1}{\mathrm{j}\omega C}}{R_1 \cdot R_2 + \frac{R_2}{\mathrm{j}\omega C} + \frac{R_1}{\mathrm{j}\omega C}} = \boxed{\frac{R_1}{R_1 + R_2 + \mathrm{j}\omega C R_1 R_2}} \qquad \hat{=} \frac{a + jb}{c + jd} \qquad a,b,c,d \in \mathbb{R}$$

$$= \boxed{\frac{R_1(R_1+R_2) - \mathrm{j}\omega C R_1^2 R_2}{(R_1+R_2)^2 + (\omega C R_1 R_2)^2}} \quad \text{(kartesisch)} \qquad \hat{=} \frac{a' + jb'}{c'} \qquad a',b',c' \in \mathbb{R}$$

Betrag: $\quad |A(\omega)| = \dfrac{R_1}{\sqrt{(R_1+R_2)^2 + (\omega C R_1 R_2)^2}} \qquad$ with $\quad |\underline{F}(\mathrm{j}\omega)| = \dfrac{\sqrt{a^2+b^2}}{\sqrt{c^2+d^2}}$

Phase: $\quad \varphi(\omega) = \arctan\left(\dfrac{-\omega C R_1^2 R_2}{R_1(R_1+R_2)}\right) \qquad$ with $\quad \varphi(\omega) = \arctan\left(\dfrac{b'}{a'}\right)$

$$= \arctan\left(\frac{-\omega C R_1 R_2}{R_1 + R_2}\right)$$

Altern.: $\quad \varphi(\omega) = \arctan(0) - \arctan\left(\dfrac{\omega C R_1 R_2}{R_1 + R_2}\right) \qquad$ with $\quad \varphi(\omega) = \overbrace{\varphi_Z}^{\arctan(\frac{b}{a})} - \overbrace{\varphi_N}^{\arctan(\frac{d}{c})}$

Solution b) Limit behaviour and sketch

$A(\omega) = \frac{R_1}{\sqrt{(R_1+R_2)^2+(\omega C R_1 R_2)^2}}$

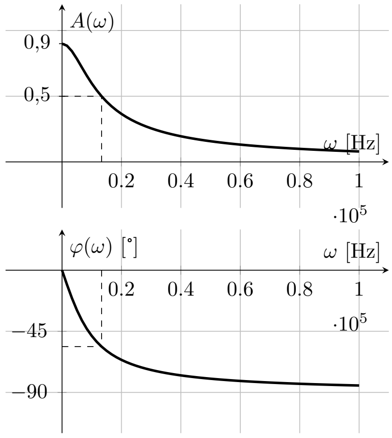
$\lim\limits_{\omega\to 0} A(\omega) = \frac{R_1}{R_1+R_2} = 0{,}9$

$\lim\limits_{\omega\to\infty} A(\omega) = \frac{R_1}{\infty} = 0$

$\varphi(\omega) = \arctan(-\frac{\omega C R_1 R_2}{R_1+R_2})$

$\lim\limits_{\omega\to 0} \varphi(\omega) = \arctan(0) = 0$

$\lim\limits_{\omega\to\infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$



Solution c) Determine specific operating point

Total: Frequency f_1 where $\hat{U}_2 = \frac{1}{2}\hat{U}_1$ and phase shift u_2 relative to u_1 at f_1:

$$A(\omega_1) = \frac{R_1}{\sqrt{(R_1+R_2)^2+(\omega C R_1 R_2)^2}} \overset{!}{=} \frac{1}{2} \quad \text{with} \quad A(\omega) = \frac{U_2}{U_1} = \frac{\hat{U}_2}{\hat{U}_1} \overset{!}{=} \frac{1}{2}\Big|_{f=f_1}$$

$$\Rightarrow \frac{(R_1+R_2)^2+(\omega_1 C R_1 R_2)^2}{R_1^2} = 4 \quad \Rightarrow \quad (R_1+R_2)^2+(\omega_1 C R_1 R_\mathrm{C})^2 = 4R_1^2$$

$$\Rightarrow (\omega_1 C R_1 R_2)^2 = 4R_1^2 - (R_1+R_2)^2 \quad \Rightarrow \quad \omega_1 = \pm\frac{\sqrt{4R_1^2-(R_1^2+2R_1R_2+R_2^2)}}{C R_1 R_2}$$

$$\omega_1 = 1{,}3304\cdot 10^4\ \frac{1}{\mathrm{s}},\ f_1 = 2{,}117\ \mathrm{kHz} \quad \text{with} \quad \omega_1 < 0 \text{ unphysikalisch},\ f = \frac{\omega}{2\pi}$$

$$\varphi(\omega_1) = \arctan(-1{,}4967) = -56{,}25^\circ \quad \text{with} \quad \varphi(\omega) = \arctan\left(\frac{\mathrm{Im}\{\underline{F}\}}{\mathrm{Re}\{\underline{F}\}}\right)$$

3 Logarithmic representation, frequency response, cut-off frequency

Logarithmic scales are often used in electrical engineering and communications engineering to represent the frequency response of filters and amplifiers. The logarithmic representation provides a better overview of the behaviour of the system across the entire frequency range.

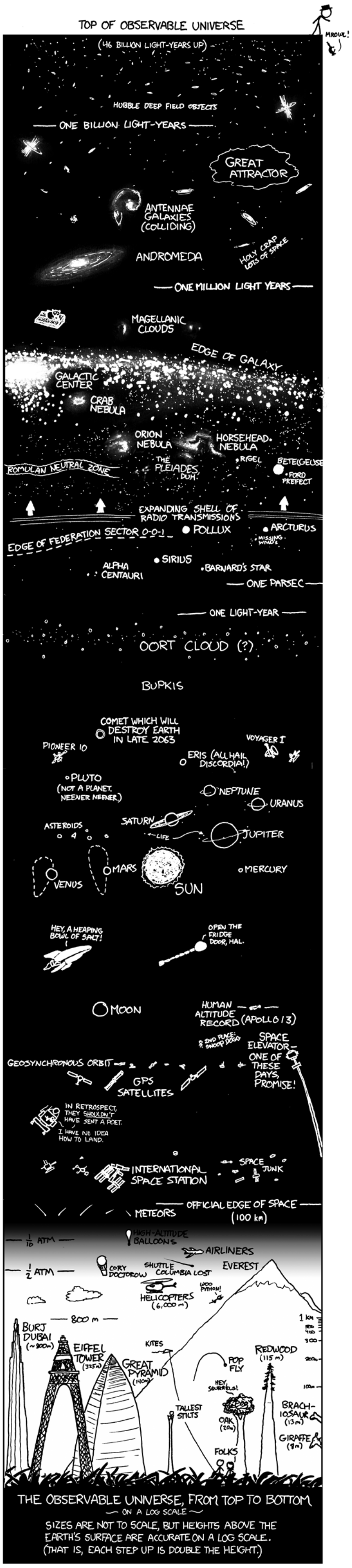
Figure 3.1 shows two comic strips from XKCD that humorously illustrate the advantages of logarithmic representation.

In this chapter, we will look at the logarithmic representation of frequency responses in the form of so-called Bode diagrams. To describe the frequency response of simple filter circuits, the unit decibel, as well as the cut-off frequency and the order of filters, are defined. In addition, bandpass and band-stop filters are introduced.

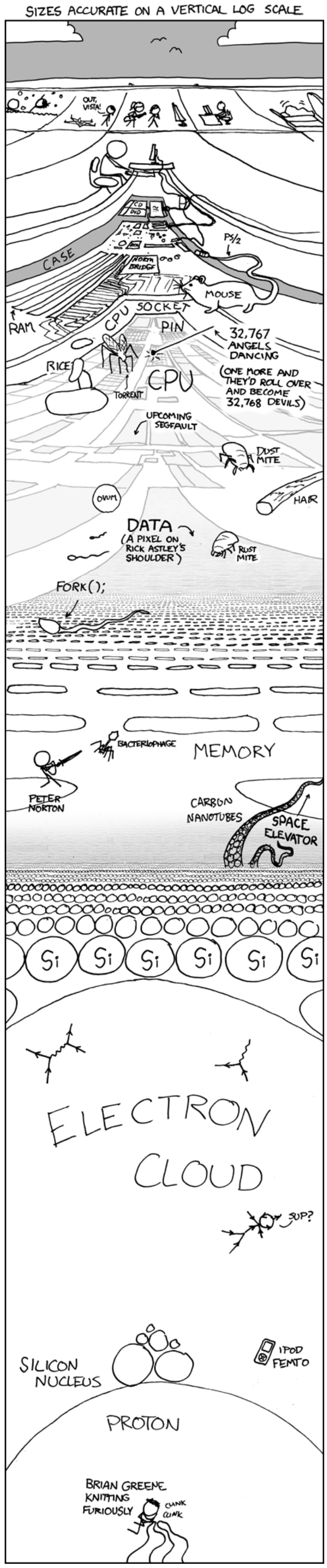
Learning objectives: Logarithmische Darstellung, Frequenzgang

Students learn:

- Know the differences between linear and logarithmic representation.
- Know and use decibels as a unit of measurement.
- Know and determine the cut-off frequencies of filter circuits.
- Construct and interpret Bode diagrams.
- Know and understand how bandpass and band-stop filters work.
- Know and understand the structure and behaviour of higher-order filters.



(a) Comic *Height*, Log-Skala Height[1, #482]



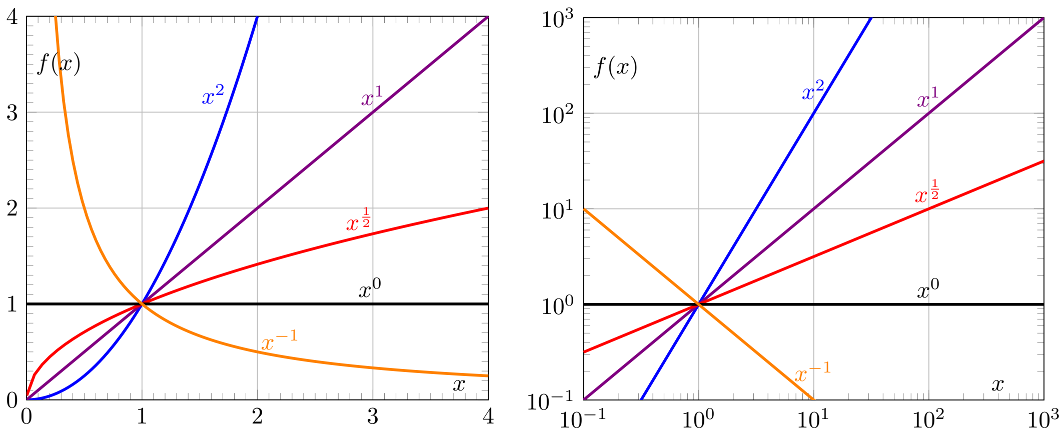
(b) Comic *Depth*, Log-Skala Depth[1, #485]

Figure 3.1: xkcd comics on logarithmic representations of quantities

3.1 Logarithmic representation of power functions

In logarithmic representation, a logarithmic scale is used for one or more axes. A linear increase in the values to be plotted corresponds to a logarithmic increase in the distance on the scale. Conversely, this can be illustrated more clearly as follows: With a linear increase in the distance on the scale, the power of the values to be plotted increases linearly by the same amount.

Logarithmic scales are generally useful for representing value ranges across several orders of magnitude. A special feature of the double logarithmic representation is that power functions are represented as straight lines. Figure 3.2 shows a comparison of several power functions in linear representation (left) and in double logarithmic representation with base 10 (right).



(a) Power functions, linear axes

(b) Power functions, logarithmic axes

Figure 3.2: Comparison of linear and logarithmic scaling of power functions

In double logarithmic representation, the exponent of a power function corresponds to the slope of the straight line. A shift in the y-direction corresponds to a pre-factor a for the function $f(x)$, a shift in the x-direction corresponds to a pre-factor x_{off} for x in the function. These properties make logarithmic representation particularly suitable for displaying amplitude responses.

3.2 Definition of decibel

Decibel is an auxiliary unit of measurement used to indicate the decimal logarithmic ratio of two quantities. It is used in signal theory and communications engineering, for example to indicate the amplification/attenuation of a component or a signal path. One decibel [dB] corresponds to ten times the base unit bel [B].

The bel is defined as the designation of the decimal logarithmic ratio (symbol Q) of two power quantities (index P) with the same unit as shown in equation 3.1. In connection with voltage or current as so-called power root quantities (formerly field quantities, index F), the (deci-)bel can also be used in linear systems as shown in equation 3.2. The conversion is based on the proportionality of $P \sim U^2$ and $P \sim I^2$, respectively, which results in the factor 2.

Power indicators (e.g.P, W) $$Q_{(\mathrm{P})} = \log \frac{P_2}{P_1}\ \mathrm{B} = 10 \cdot \log \frac{P_1}{P_2}\ \mathrm{dB} \tag{3.1}$$

Power root group(e.g.U, I) $$Q_{(\mathrm{R})} = \log \frac{U_2^2}{U_1^2}\ \mathrm{B} = 20 \cdot \log \frac{U_1}{U_2}\ \mathrm{dB} \tag{3.2}$$

The value Q in dB describes the amplification of a system from the input signal (index 1) to the output signal (index 2). Table 3.1 shows the corresponding voltage ratios $\frac{U_2}{U_1}$ and power ratios $\frac{P_2}{P_1}$ for typical decibel values. A change of 6 dB corresponds, rounded to the second decimal place, exactly to the factor 2 for voltages and, rounded to the first decimal place, exactly to the factor 4 for powers.

Table 3.1: Typical decibel values (amplification in dB) for voltage and power ratios

Pow.ro.con.	$\frac{U_2}{U_1}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	2	$\sqrt{10}$	10	20	100
Power indicators	$\frac{P_2}{P_1}$	$\frac{1}{2}$	1	2	4	10	100	400	10.000
Amplification	Q_{dB}	−3 dB	0 dB	+3 dB	+6 dB	10 dB	20 dB	26 dB	40 dB

Since 3 dB corresponds very precisely to the factor 2 for power (factor $\sqrt{2}$ for voltage) and 6 dB to the factor 4, estimates can be made relatively easily. Compared to the SI unit neper, which is based on the natural logarithm, the decibel has therefore become established in practice as a logarithmic auxiliary unit of measurement.

Faustformeln für Spannungsverhältnisse:

- Factor $1 = 0\,\text{dB}$
- Factor $2 \approx 6\,\text{dB}$
- Factor $4 \approx 12\,\text{dB}$ with $4 = 2^2 \approx 2 \cdot 6\,\text{dB}$
- Factor $5 \approx 14\,\text{dB}$ with $5 = 10 \cdot \frac{1}{2} \approx 20\,\text{dB} - 6\,\text{dB}$
- Factor $8 \approx 18\,\text{dB}$ with $8 = 2^3 \approx 3 \cdot 6\,\text{dB}$
- Factor $10 = 20\,\text{dB}$ $\Longrightarrow$ $20\,\text{dB}\,/\text{Dek}$ (voltage-related)

As an example, Figure 3.4 shows the logarithmic y-axis for $A(\omega)$ once without units (left) and once in dB (right) with the integer factors or decibel values from the above rule of thumb labelled.

Phenologically, the following applies:

- There is amplification for $\frac{U_2}{U_1} > 1$
- Attenuation occurs for $\frac{U_2}{U_1} < 1$

For amplification and damping as physical quantities, the following applies: they correspond to each other in dB with opposite signs. The usual symbols are G (from 'gain') for the amplification of components and D (from 'damping') for their attenuation. Both values are typically given in dB and, unless otherwise specified, in relation to voltage levels. In data sheets, attenuation is sometimes also given in dB with a negative sign.

3.3 Cut-off frequency of simple filter circuits

High-pass and low-pass filters allow input signals to be filtered depending on their frequency, as shown in Chapter 2.3 for a low-pass filter and in Chapter 2.5 for a high-pass filter.

The frequency range of the amplitude responses of both filters is divided into a **passband** and a **stopband**. The so-called **cutoff frequency** f_g or limit (circuit) frequency ω_g serves as the boundary. ω_g is defined by a fixed ratio of effective output voltage to input voltage.

In the case of simple filter circuits, the following applies:

$$\left.\frac{U_2}{U_1}\right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}} \tag{3.3}$$

According to Table 3.1, this corresponds to an attenuation of 3 dB or an amplification of −3 dB. The output power at ω_g is half the input power.

Example: First-order low-pass filter

Using equation 2.7, the following applies to the first-order RC low-pass filter for the cut-off frequency:

$$\begin{aligned} A(\omega) &= \frac{1}{\sqrt{1+(\omega CR)^2}} \\ A(\omega_g) &= \frac{1}{\sqrt{1+(\omega_g CR)^2}} \overset{!}{=} \frac{1}{\sqrt{2}} \\ \ldots \quad \Longrightarrow \quad \omega_g &= \frac{1}{CR} \end{aligned} \tag{3.4}$$

The cut-off frequency ω_g corresponds to the reciprocal of the time constant $\tau = CR$ of the RC element.

The frequency, amplitude and phase responses of the first-order RC low-pass filter from equations 2.6, 2.8 and 2.7 an be normalised to ω_g. This results in the following component-independent values:

$$F(\mathrm{j}\omega) \overset{2.6}{=} \frac{1}{1+\mathrm{j}\omega CR} \quad \overset{3.4}{\Longrightarrow} \quad F(\mathrm{j}\omega) = \frac{1}{1+\mathrm{j}\,(\omega/\omega_\mathrm{g})} \tag{3.5}$$

$$A(\omega) \overset{2.7}{=} \frac{1}{\sqrt{1+(\omega CR)^2}} \quad \overset{3.4}{\Longrightarrow} \quad A(\omega) = \frac{1}{\sqrt{1+(\omega/\omega_\mathrm{g})^2}} \tag{3.6}$$

$$\varphi(\omega) \overset{2.8}{=} -\arctan(\omega CR) \quad \overset{3.4}{\Longrightarrow} \quad \varphi(\omega) = -\arctan(\omega/\omega_\mathrm{g}) \tag{3.7}$$

Figure 3.3 shows the amplitude response as a function of ω/ω_g. In addition to the actual curve in red, the curve of an idealised low-pass filter with high slope is also shown in blue.

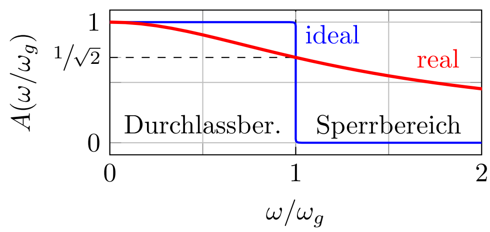


Figure 3.3: Passband and stopband of a simple low-pass filter, linear scale

A change in the cut-off frequency results in a compression or stretching of the amplitude response along the frequency axis in a linear representation.

Example: First-order high-pass filter

Similarly, we can determine the cut-off frequency for the first-order RC high-pass filter:

$$\begin{aligned} A(\omega_g) &= \frac{1}{\sqrt{1+\left(\frac{1}{\omega_g CR}\right)^2}} \overset{!}{=} \frac{1}{\sqrt{2}} \\ \ldots \quad \Longrightarrow \quad \omega_g &= \frac{1}{CR} \end{aligned} \tag{3.8}$$

The cut-off frequency of both filters is therefore identical for the same component values R and C. The frequency response of the high-pass filter is analogous to that of the low-pass variant:

$$F(\mathrm{j}\omega) = \frac{1}{1 - \mathrm{j}\left(\omega_\mathrm{g}/\omega\right)} \tag{3.9}$$

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\omega_\mathrm{g}/\omega\right)^2}} \tag{3.10}$$

$$\varphi(\omega) = \arctan\left(\omega_\mathrm{g}/\omega\right) \tag{3.11}$$

3.4 First-order low-pass filter

Frequency responses are typically represented in **Bode diagrams**. Bode diagrams consist of a double logarithmic representation of the amplitude response and a single logarithmic representation of the phase response. The x-axis for the (circular) frequency is logarithmically scaled in both representations. The y-axis is logarithmically scaled for the amplitude response and linearly scaled for the phase response.

Figure 3.4 shows the amplitude response from Eq. 3.6 of a first-order lowpass filter as a Bode diagram. The passband for $\omega < \omega_g$ and the stopband for $\omega > \omega_g$ are marked, as is the position for $\omega = \omega_g$. The frequency specification on the x-axis has been normalised to the cut-off frequency and is therefore unitless.

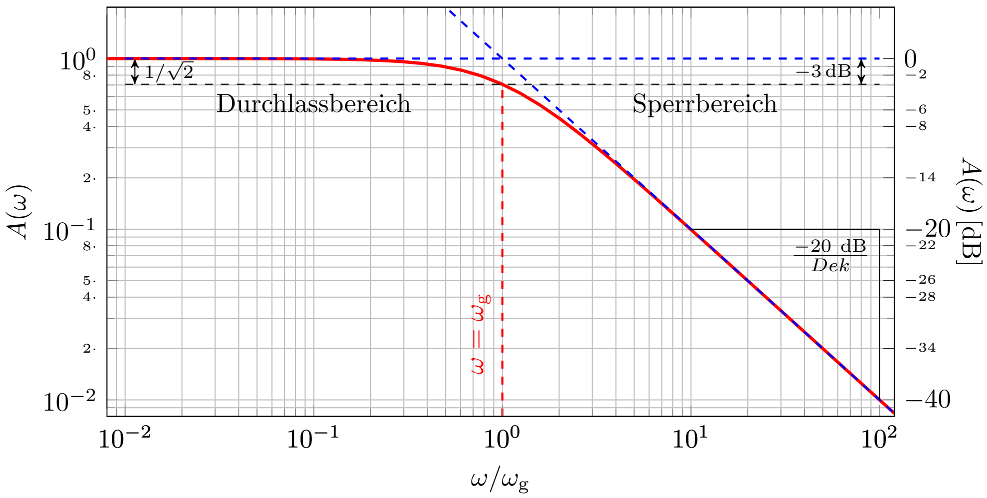


Figure 3.4: Bode diagram, first-order low-pass filter, amplitude response

The division of the frequency range into passband and stopband by the cut-off frequency is clearly visible. The amplitude response curve can be approximated in both ranges using asymptotes (straight lines) [blue dotted lines]:

$$\lim_{\omega\to 0} A(\omega) = 1 \quad \Longrightarrow \quad A(\omega) \approx 1 \quad \text{für} \quad \omega < \omega_g \quad \text{(Passband)}$$

$$\lim_{\omega\to \infty} A(\omega) = \frac{\omega_g}{\omega} \quad \Longrightarrow \quad A(\omega) \approx \frac{\omega_g}{\omega} \quad \text{für} \quad \omega > \omega_g \quad \text{(Stopband)}$$

The slope of the asymptote (straight line) in the cut-off range is $-20dB/dec$, due to the proportionality of $A(\omega) \sim \omega^{-1}$. The approximation by both asymptotes deviates by a maximum of 3 dB for

$\omega = \omega_g$ from the actual curve. At the cut-off frequency, both asymptotes intersect, which is not generally true for filters. Both asymptotes (straight lines) and the point $A(\omega_g) = 1/\sqrt{2}$ were suitable for constructing a sketch. For the sketch, an arc of asymptote is drawn from $A(\omega_g) = 1/\sqrt{2}$ to asymptote in the transition range of factor five greater or less than the cut-off frequency ($\frac{1}{5}\,\omega_g < \omega < 5\,\omega_g$) .

A change in the cut-off frequency causes a shift along the frequency axis in the Bode diagram. This applies to both the amplitude and phase response, as the frequency axis is logarithmically scaled in both representations. Normalisation to the cut-off frequency is useful for representing the functional behaviour of the frequency response independently of specific component values.

Figure 3.5 shows the corresponding phase response as a Bode diagram, also with the frequency normalised to the cut-off frequency. The curve shape of the phase response resembles the curve shape of an arctangent in linear representation with corresponding displacement and compression.

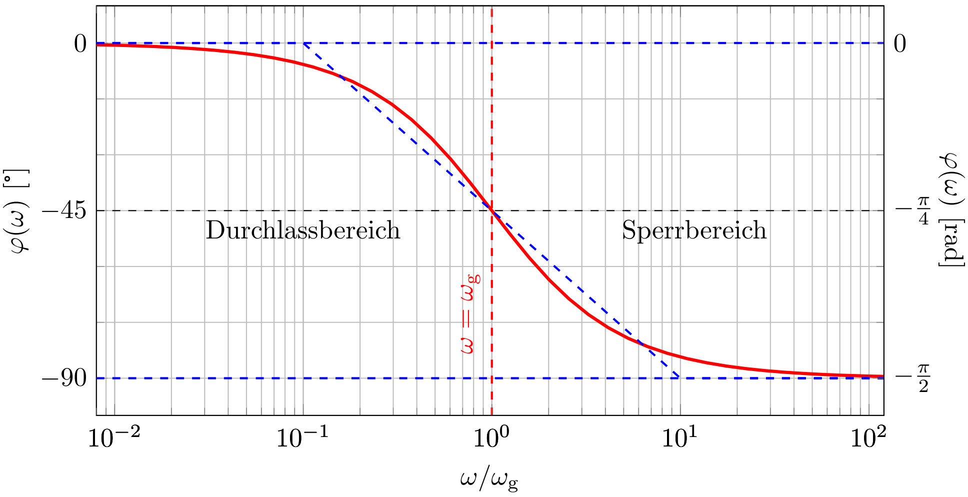


Figure 3.5: Bode diagram, first-order low-pass filter, phase response

The range of values of the phase response is determined by the two horizontal asymptotes. $\varphi(\omega) = 0\,^\circ$ für $\omega \to 0$ und $\varphi(\omega) = -90\,^\circ$ für $\omega \to \infty$ begrenzt. In this representation, the phase curve has point symmetry at the inflection point in $\varphi(\omega_g) = -45\,^\circ$ and no extrema.

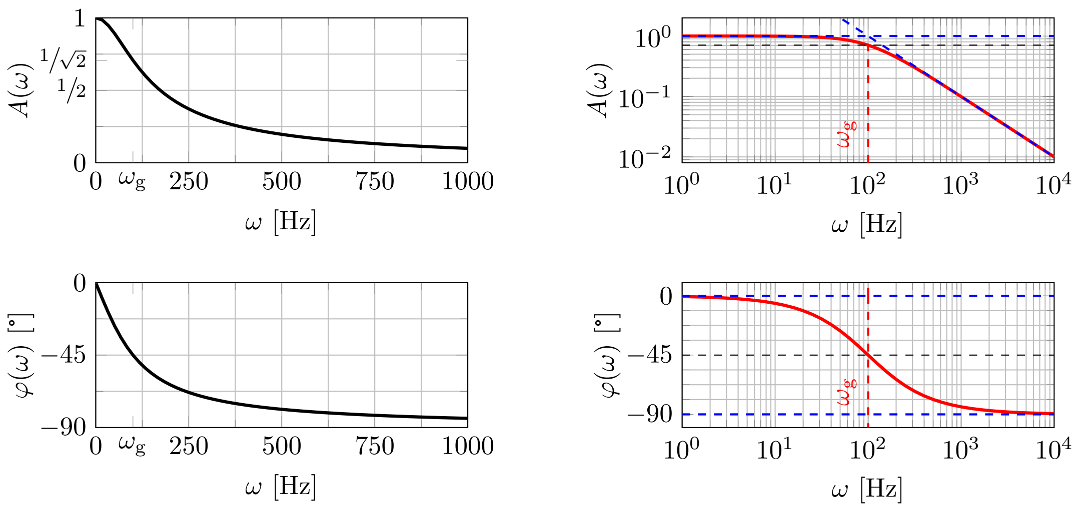
Let ω_n be the normalised frequency ω/ω_g. The phase response can then be approximated in sections by the following three straight lines:

$$\varphi(\omega) \approx \begin{cases} 0\,^\circ & \text{für} \quad \omega_n < 10^{-1} \quad \text{Passband (without Cutoff frequency)} \\ -45\,^\circ + \dfrac{-45\,^\circ}{\text{Dek}}\omega_n & \text{für} \quad 10^{-1} \leq \omega_n \leq 10 \quad \text{transition band} \\ -90\,^\circ & \text{für} \quad 10 < \omega_n \quad \text{Stopband (without Transition band)} \end{cases} \tag{3.12}$$

The maximum deviation of the approximation is $\pm\ 5.71^\circ$ at the inflection points at $\omega_g \pm 1$ dec.

3.4.1 Comparison of linear and logarithmic representations

For comparison, Figure 3.6 shows the amplitude and phase response of a first-order lowpass filter in linear and logarithmic representation (Bode diagram).



(a) Linear representation

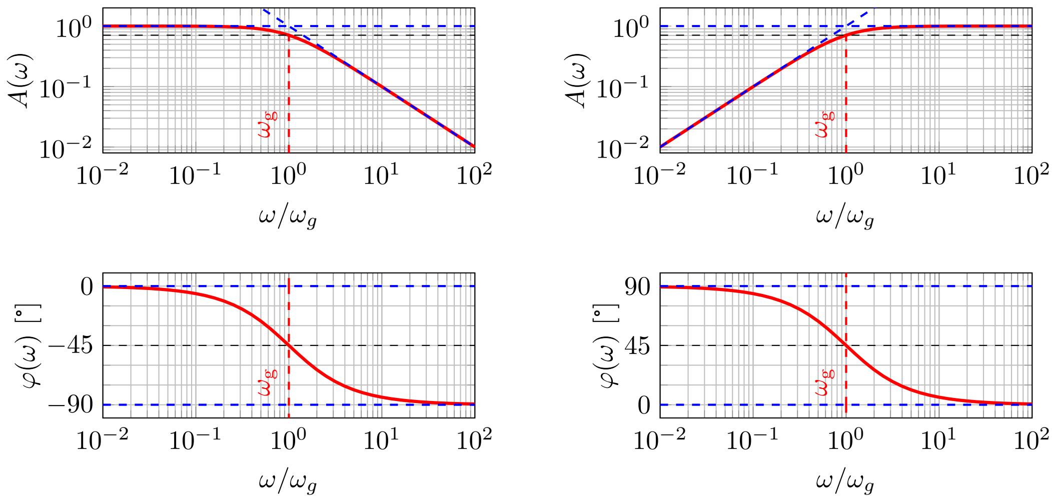
(b) Logarithmic representation

Figure 3.6: Comparison of linear and logarithmic representation of the first-order low-pass filter

Unlike linear representation, in logarithmic representation the derivative of a function does not correspond to the readable slope in the curve. This can be clearly seen in Figure 3.6. The slope of $A(\omega)$ in linear representation approaches 0 approaches 0 for $\omega \to 0$ as well as for $\omega \to \infty$. [See chapter 2.3] In the Bode diagram, the slope for $\omega \to 0$ is also zero, but approaches approaches a minimum slope of -20 dB /dec for $\omega \to \infty$. In the phase response, the slope (derivative) for $\omega \to 0$ is minimal and negative in linear representation. However, in logarithmic representation, $\varphi(\omega)$ appears flattest (slope towards 0) for $\omega \to 0$.

3.4.2 Comparison of first-order low-pass and high-pass filters

Figure 3.7 shows the frequency responses of a lowpass and a highpass 1st order filter in a Bode diagram for comparison. The frequency range (x-axis) is normalised to the respective cut-off frequency in both cases.



(a) First-order low-pass filter

(b) First-order high-pass filter

Figure 3.7: Comparison of first-order low-pass and high-pass filters, Bode diagrams

3.5 Bandpass, Bandstop

Bandpass filters and band-stop filters are filter types used to filter frequency bands.

Similar to low-pass and high-pass filters, both are widely used in signal processing, for example in the fields of audio, communications, measurement and control technology. Other applications can be found, for example, in energy technology when feeding energy into the electrical supply network to ensure a stable mains frequency.

Bandpass filters filter out signals outside a specific frequency band, while band-stop filters filter out signals within a specific frequency band. The frequency bands are defined by an upper and lower cut-off frequency f_{go} and f_{gu}.

$$\left.\frac{\mathrm{U}_2}{\mathrm{U}_1}\right|_{f=f_{\mathrm{go}}} = \frac{1}{\sqrt{2}} \quad \text{und} \quad \left.\frac{\mathrm{U}_2}{\mathrm{U}_1}\right|_{f=f_{\mathrm{gu}}} = \frac{1}{\sqrt{2}} \tag{3.13}$$

Derived quantities are the **bandwidth** B and the **centre frequency** f_{M}. The bandwidth is defined as the difference between the upper and lower cut-off frequencies:

$$B = \Delta f = f_{\mathrm{go}} - f_{\mathrm{gu}} \tag{3.14}$$

The centre frequency f_{M} is defined as the geometric mean of both cut-off frequencies:

$$f_{\mathrm{M}} = \sqrt{f_{\mathrm{go}} \cdot f_{\mathrm{gu}}} \tag{3.15}$$

In logarithmic representation, f_{M} is, by definition, exactly in the middle between both cut-off frequencies. This allows the size and position of the frequency band in the frequency spectrum to be described using B and f_{M}.

3.5.1 Implementation: Series or parallel connection of low-pass and high-pass filters

Bandpass and band-stop filters can be implemented by combining a low-pass and high-pass filter. This provides a clear illustration of how combined filter circuits behave.

Connecting a low-pass and high-pass filter in series creates a band-pass filter, provided that the cut-off frequencies are selected appropriately. For band-pass behaviour, the upper cut-off frequency must be realised by the low-pass filter and the lower cut-off frequency by the high-pass filter. Mathematically, the resulting frequency response can be described as the product of the individual frequency responses with $\underline{F}_{\mathrm{BP}} = \underline{F}_{\mathrm{TP}} \cdot \underline{F}_{\mathrm{HP}}$. A prerequisite for multiplying the individual frequency responses to obtain the overall frequency response is that the downstream four-pole circuit has no feedback on the output signal of the upstream four-pole circuit.

Figure 3.8 shows an example of the circuit diagram of such a second-order bandpass as a series connection of an RC low-pass filter and an RC high-pass filter, both of the first order. The symbolic amplifier block between the two filters prevents the high-pass filter from having a feedback effect on the output voltage of the low-pass filter. Only under the assumption of no feedback can the frequency responses of the low-pass and high-pass filters be multiplied by connecting them in series.

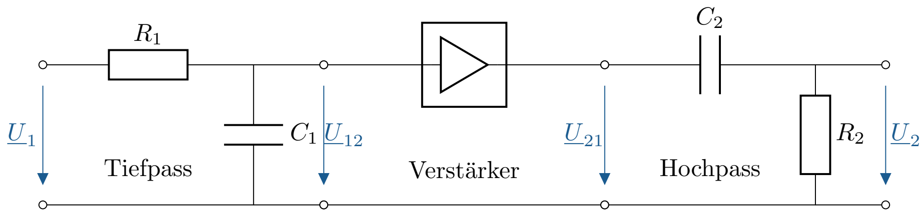


Abbildung 3.8: Bandpass 2. Ordnung, Schaltbild

A band-stop filter can be implemented by connecting the inputs of the low-pass and high-pass filters in parallel and their outputs in series. This ensures that both individual filters receive the same input voltage and the output voltages are added together. The resulting frequency response is the sum of the individual frequency responses with $\underline{F}_{\mathrm{BS}} = \underline{F}_{\mathrm{TP}} + \underline{F}_{\mathrm{HP}}$. In this case, the upper cut-off frequency must be implemented by the high-pass filter and the lower cut-off frequency by the low-pass filter.

3.5.2 Bode diagram

Figure 3.9 shows the Bode diagram of the bandpass from Fig. 3.8. The amplitude response and phase response are superimposed on a common X-axis for better comparison of the two graphs. The gain factor is 1 (passive) and the cut-off frequencies are 10^2Hz and 10^4Hz.

The upper and lower cut-off frequencies ω_{go} and ω_{gu} are marked in both representations (red dashed line, vertical), as is the -3 dB limit (black dashed line, horizontal) whose intersection with $A(\omega)$ defines the cut-off frequencies.

The bandwidth is indicated in the amplitude response as a double arrow between the two cut-off frequencies. Due to the logarithmic scaling, the bandwidth does not correspond to the geometric distance shown in the diagram.

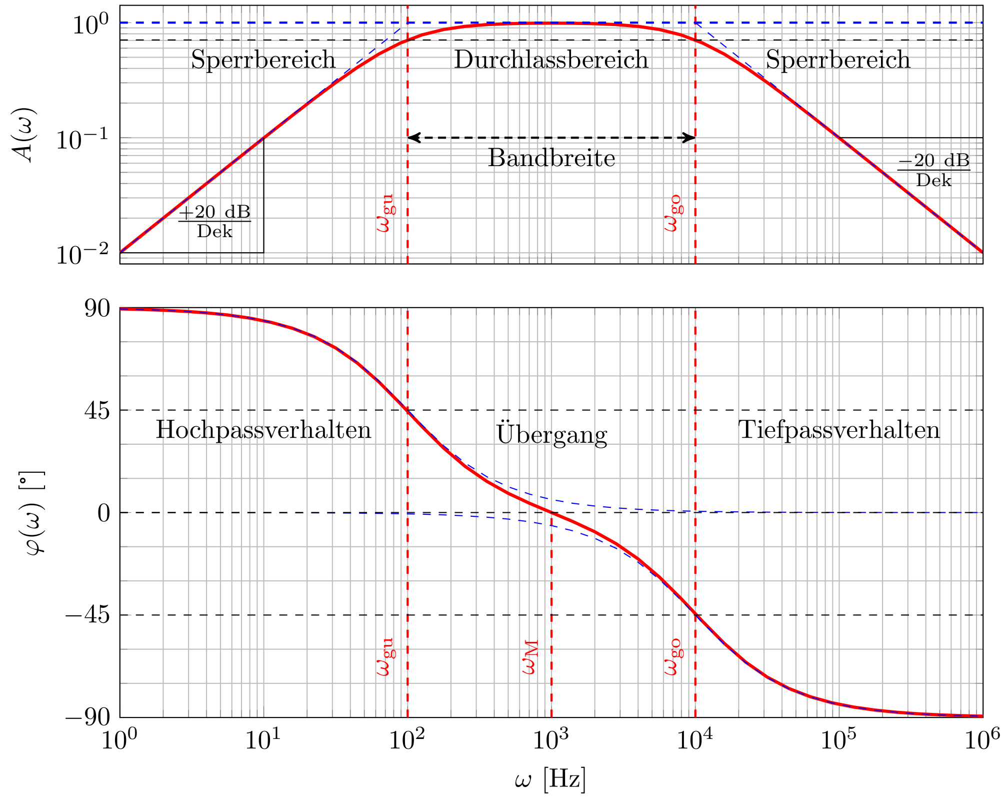


Figure 3.9: Second-order bandpass, Bode diagram

The diagram shows:

$$\omega_{gu} = 10^2\text{Hz} \qquad \text{und} \qquad \omega_{go} = 10^4\text{Hz} \tag{3.16}$$

$$B \overset{3.14}{=} \Delta f = \frac{\omega_{go} - \omega_{gu}}{2\pi} = \frac{9900}{2\pi}\text{Hz} \tag{3.17}$$

$$f_m \overset{3.15}{=} \frac{\omega_m}{2\pi} = \frac{\sqrt{\omega_{go} \cdot \omega_{gu}}}{2\pi} = \frac{10^3}{2\pi}\text{Hz} \tag{3.18}$$

The centre frequency ω_m is marked in the phase response. The phase shift there is 0°, since the phase shifts of the low-pass and high-pass (blue dashed line, thin) cancel each other out at this point.

The high-pass behaviour in the lower cut-off range (with +20 dB /Dek and $\varphi(\omega) > 0°$) and the low-pass behaviour in the upper cut-off range (with −20 dB /Dek and $\varphi(\omega) > 0°$) are clearly visible.

3.6 Order of filters, example filter of higher order

Figure 3.10 shows an example comparison of the amplitude responses of various higher-order filters. A Butterworth, Chebyshev, and Cauer filter are shown in comparison with an ideal low-pass filter.

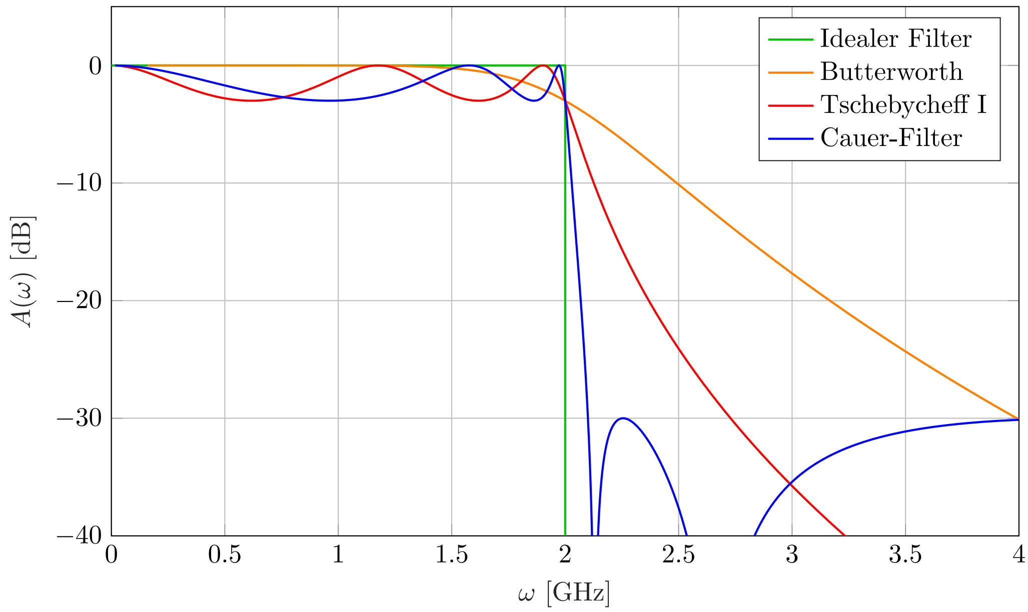


Figure 3.10: Example: Higher-order filter

Explain the order based on the total steepness of the flanks.

- Low-pass/high-pass (one edge): Order n at $\pm\, n \cdot 20\,\text{dB}\,/\text{dec}$
- Band-pass/band-stop (two edges): Order $2n$ at $\pm\, n \cdot 20\,\text{dB}\,/\text{dec}$

Second-order filter through: RLC or combination of two first-order filters: 2x (RC) or 2x (RL) (first order)

Filters of higher order than 2 are possible by cascading lower-order filters.

Compare: Mexle Wiki

Key point:

$n \cdot 20$ dB/Dek Slope for n-th order filter for all edges summed.
Higher order possible by cascading lower order filters.

4 Location curve

The representation of locus curves is a means of visualising the parameter-dependent change in complex variables. Impedance and admittance locus curves are useful for analysing the frequency-dependent behaviour of single-port networks. These curves can be used to represent the change in impedance or admittance as a function of (circular) frequency.

Learning objectives: Locus curve

Students learn:

- Know locus curves and possible areas of application.
- Construct and interpret locus curves in principle.
- Know impedance and admittance locus curves of basic circuits.

4.1 Definition of location curve

A locus curve is the graphical representation of a complex quantity $\underline{z}$ as a function of a real parameter p in the complex plane[2, See]:

$$\underline{z}(p) = \mathrm{Re}\{\underline{z}(p)\} + \mathrm{j}\,\mathrm{Im}\{\underline{z}(p)\} \quad \text{mit } \underline{z} \in \mathbb{C};\ p \in \mathbb{R} \tag{4.1}$$

Locus curves are used in many different areas to visualise possible states of a system. Figure 4.1 shows an example of the current locus curve of an asynchronous machine in the complex plane, also known as the Heyland circle or Ossanna circle. [SOURCE:XYZ] The complex stator current $\underline{I}_s$ can be seen as a function of the slip s. This form of representation is suitable, for example, for visualising different operating states. In the case of the asynchronous machine, these are motor, brake and generator operation.

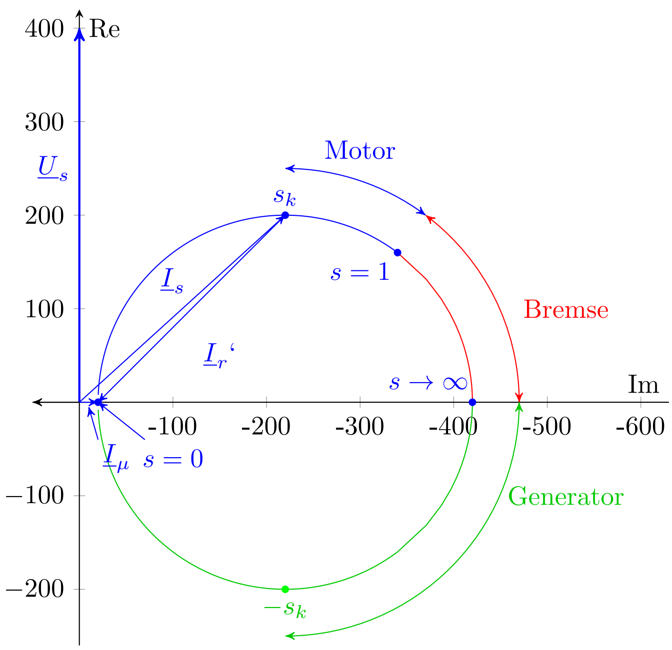


Figure 4.1: Example of the current-frequency curve of an asynchronous machine with operating states

In the example of the current locus curve of an asynchronous machine, various power values can also be determined graphically as shown in Figure 4.2. The examples serve to illustrate the definition and possible areas of application of locus curves, which is why we will refrain from deriving both examples.

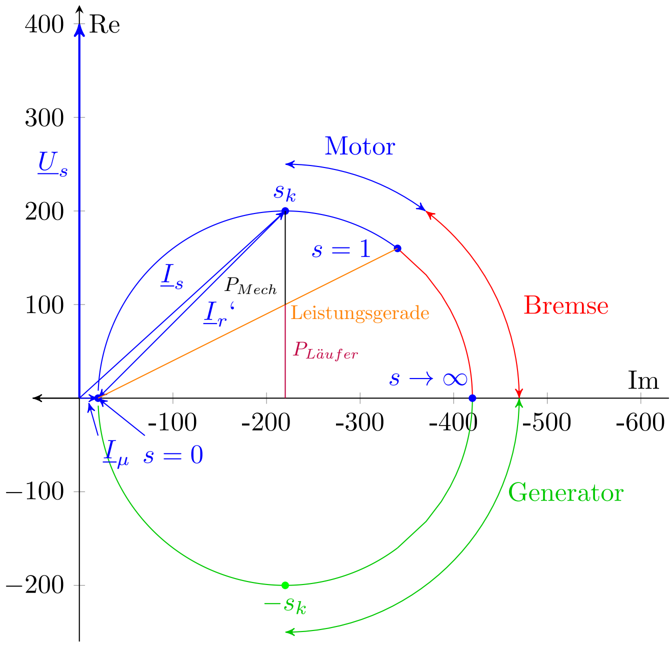


Figure 4.2: Example of the current-frequency curve of an asynchronous machine with power variables

Locus curves are well suited for visualising system states, variables and their influence on each other.

4.2 Relationship between pointer diagram and location curve

A pointer in the phasor diagram represents a complex quantity stationary in the complex plane. This form of representation serves to visualise phase shifts and magnitude ratios. Typical quantities for alternating current circuits include voltage, current, power and impedance.

Locus curves describe the path that a pointer travels when a parameter varies. The locus curve can thus be understood as a generalisation of a pointer in the pointer diagram. [2, Compare]Common parameters are frequency and component quantities. Common quantities are impedance and admittance.

Highlighting individual pointers with an indication of the varying parameter can clarify the course of the locus curve and facilitate interpretation. The rate and direction of change in position provide information about the dynamics of the system.

Key point: Location curve

Displays size in a complex plane as a curve (set of points) depending on parameters.

4.3 impedance locus curve

Impedance locus curves show the impedance of a system as a function of a parameter.

Example **??** shows an impedance locus curve for a variable component size.

Example **??** shows an impedance locus curve for variable frequency.

Example 4.1: Impedance locus curve RL circuit, variable resistance

Construction of the impedance locus curve of an RL series circuit for variable resistance R. The circuit is shown in Figure 4.3.

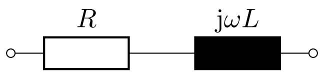


Figure 4.3: RL-Glied (Serienschaltung)

The impedance of the RL element is given by:

$$\underline{Z} = R + \mathrm{j}\omega L$$

If R is variable and described as (p-fold) multiples of a reference resistance R_0 with p in the closed interval [0,3], then the following applies:

$$R(p) = p \cdot R_0 \quad \text{mit } p \in [0{,}3]$$

Let ω and L be constant, and let the reactance ωL be equal to a reference reactance X_0:

$$\omega L = X_0 \quad \text{mit } \omega,\, L = \mathit{konst}.$$

Impedance $\underline{Z}$ as a function of p:

$$\underline{Z}(p) = p \cdot R_0 + \mathrm{j}X_0$$

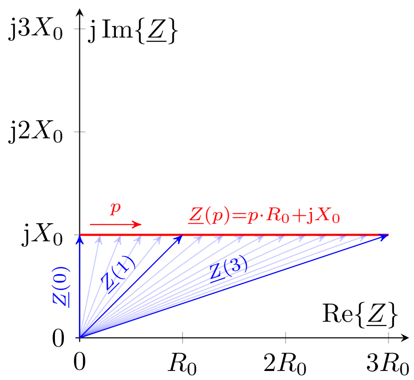


Figure 4.4: Impedance locus curve RL element (Series)

Example 4.2: Impedance locus curve RL element, variable frequency

Construction of the impedance locus curve of the RL series connection in Figure 4.5 for variable frequency f and/or variable inductance L. Setup as in Example **??**, but with variable frequency ω.

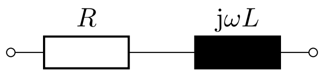


Figure 4.5: RL-Glied (Serienschaltung)

Total impedance in general:

$$\begin{aligned}\underline{Z} &= R + \mathrm{j}\omega L \\ &= R + \mathrm{j}X\end{aligned}$$

The total reactance X of the RL element is proportional to ω and proportional to L. This means that a change in ω has the same effect as a change in L on the impedance of the RL element.
With X as a multiple p of a reference reactance X_0 and $R = R_0$, the following applies:

$$\underline{Z} = R_0 + p \cdot \mathrm{j}X_0$$

Shown at a complex level:

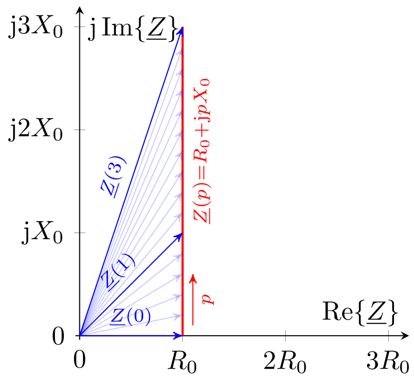


Figure 4.6: RL element (series connection)

4.4 Admittance locus curve - Inversion of locus curves

Analogous to the impedance curve, an admittance curve describes the location curve of an admittance. The admittance $\underline{Y}$ corresponds algebraically to the reciprocal of the impedance $\underline{Z}$ and vice versa:

$$\underline{Y} = \frac{1}{\underline{Z}} \quad \leftrightarrow \quad \underline{Z} = \frac{1}{\underline{Y}} \tag{4.2}$$

The conversion of impedance to admittance and vice versa is a special case of the Möbius transformation.

The following general properties of Möbius transformation therefore apply to the relationship between admittance and impedance locus curves: [3, Compare]

- Angle preservation: $\angle(\underline{Z_1},\underline{Z_2}) = \angle(\underline{Y_1},\underline{Y_2})$ with $\underline{Y}_i = \underline{Z}_i^{-1}$, $i \in [1,2]$
- Circle similarity: Circles and straight lines are mapped onto circles and straight lines

The formation of the reciprocal value corresponds in particular to inversion, an elementary type of Möbius transformation. An inversion of locus curves leads to the geometric transformations of circles and straight lines listed in Table 4.1. Figure 4.7 shows an example of the inversion of a lattice through the origin.

Table 4.1: Typical shapes in the inversion of local curves

	original local curve		inverted locus curve
Line	not through the origin	Line	not through the origin
	Not through the origin	Circle	through the origin
Circle	through the origin	Line	not through the origin
Circle	not through the origin	Circle	not through the origin

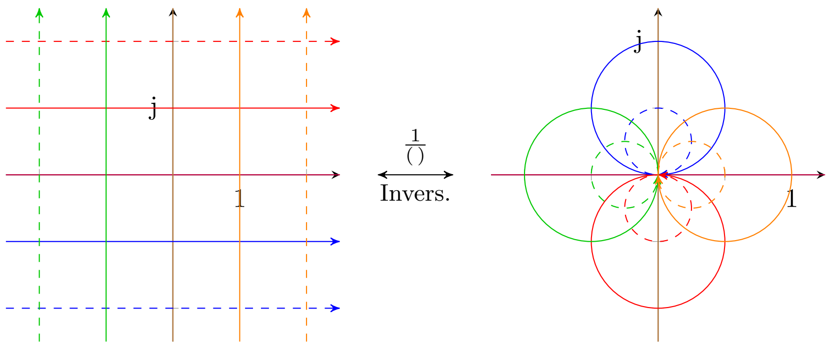


Figure 4.7: Example: Möbius transformation, inversion of lattice[1]

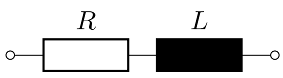
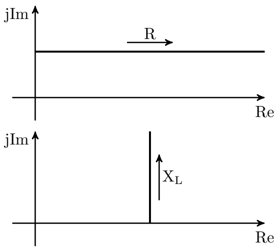
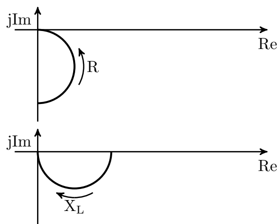
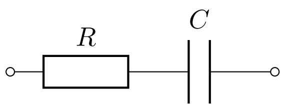
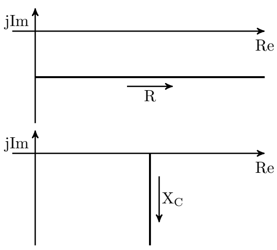
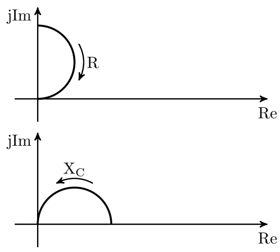
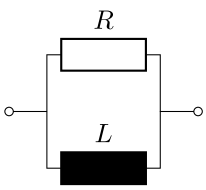
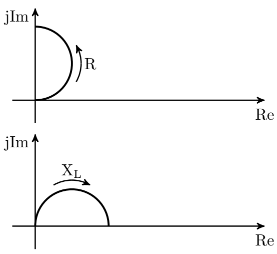
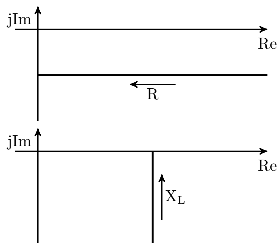
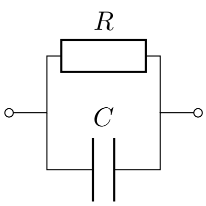
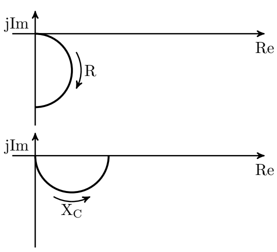
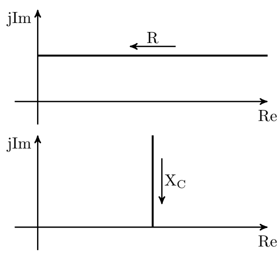
4.5 Local curves of basic circuits

Table 4.2 shows an overview of the impedance and admittance locality curves for the four basic circuits RL and RC elements in series and parallel connection.

Location curves for oscillatory LC and RLC elements differ from the location curves shown here in that the reactive component of the impedance or admittance can be either inductive or capacitive, depending on the frequency range, depending on which component predominates.

[1]Based on MoebiusInversion.svg by Chrislb, CC-BY-SA-2.0-DE, 2005, https://commons.wikimedia.org/wiki/File:MoebiusInversion.svg

Tabelle 4.2: Local curves of the basic circuits RL and RC element

Basic-circuit	**$\underline{Z}$-Locus-curve**	**$\underline{Y}$-Locus-curve**
		
		
		
		

5 Resonant circuits

Resonant circuits, also known as oscillating circuits, are oscillatory circuits that respond strongly when excited at a certain frequency (resonance). There are two types of resonant circuits: parallel resonant circuits and series resonant circuits.

A system is oscillatory if it can store energy in two forms and convert between the two forms. Resonant circuits therefore always have a capacitance (electrical storage) and an inductance (magnetic storage).

Resonant circuits are used in areas such as energy, measurement and communication technology. There, they are used, for example, to filter mains interference or signals.

Resonant circuits can, under certain circumstances, lead to undesirable effects such as extreme overvoltage or undervoltage.

Learning objectives: Resonant circuits

Students learn:

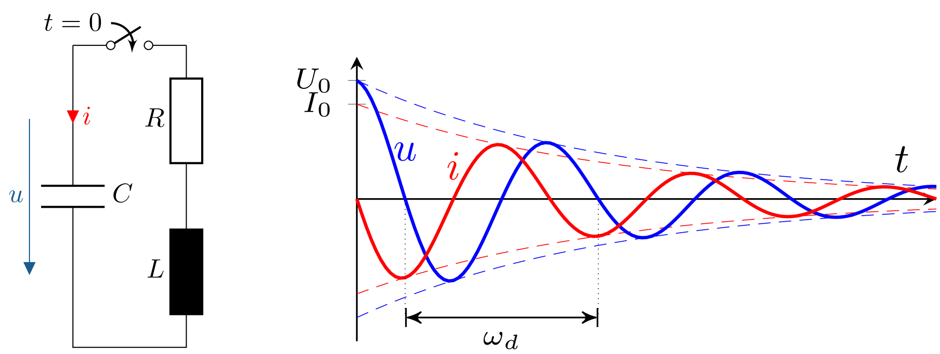
- Identifying and calculating resonance circuits
- Determine resonance frequency, characteristic impedance and quality factor
- Calculate, describe and plot resonance curves

5.1 Resonance phenomenon

Resonance occurs when oscillatory systems are excited at a frequency close to their resonant frequency.

This section first describes the state of a free oscillation and that of a forced oscillation. In subchapter 5.2, the condition for resonance is defined mathematically and calculated in the following chapter in example ?? for a series and parallel oscillating circuit.

Figure 5.1 shows an example of the circuit diagram of an RLC series resonant circuit. The series connection of R, L and C can be short-circuited via the switch shown. At time $t = 0$, the switch closes, the voltage across the capacitor is $U_0 > 0$ (capacitor charged) and no current flows $I = 0$ (inductor discharged).



(a) Circuit diagram (b) Free oscillation

Figure 5.1: RLC series resonant circuit, free oscillating

Figure 5.1b shows the time curve of the capacitive voltage u and the current i through the RLC series resonant circuit from $t = 0$. The resonant circuit is in a state of free oscillation. This means that currents and voltages oscillate without external excitation at the **natural frequency** f_d (damped) of the resonant circuit.

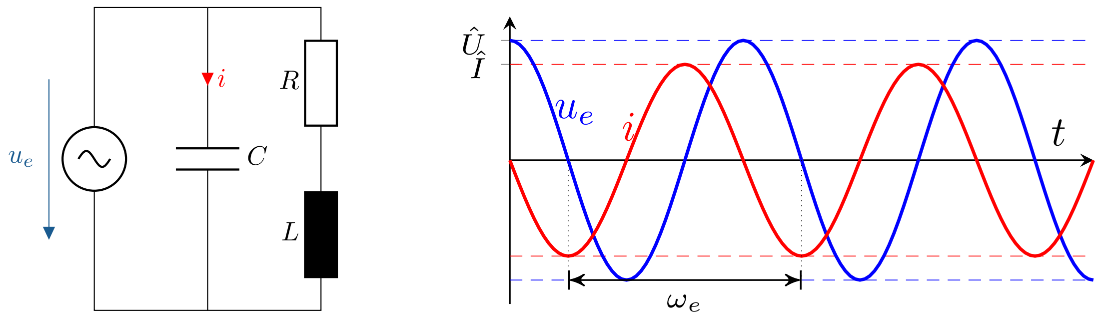
The amplitudes decay due to damping ($R > 0$). They can each be described by an exponential curve, which envelops the respective time course (dashed curve). For more precise calculations of compensation processes, see Module 12: Switching Processes.

The energy E oscillates between the electrical form in the $\underline{\vec{E}}$ field of the capacitance ($E_{el} \sim u^2$) and the magnetic form in the $\underline{\vec{H}}$ field of the inductance ($E_{mag} \sim i^2$). The minima and maxima and the zero points of both time curves show how the energy is converted back and forth twice in each period.

Due to the real (loss) power in the ohmic resistance ($R > 0$), part of the energy is converted into heat in the resistance during every energy conversion. The resistance dampens the oscillating circuit and causes the current and voltage to decay in a state of free oscillation.

The natural frequency $\omega_d = 2\pi f_d$ is independent of the amplitude. In undamped systems, it corresponds exactly to the **resonant frequency** ω_0. In slightly damped systems, ω_d is slightly smaller than ω_0. Heavily damped systems do not oscillate, but decay aperiodically.

Figure 5.2a shows an RLC series resonant circuit excited by an ideal voltage source u_e at a capacitor (forced oscillation) for comparison. Figure 5.2b shows the steady-state time response of current and voltage of the capacitance under sinusoidal excitation.



(a) Basic circuit (b) Forced oscillation

Figure 5.2: RLC series resonant circuit, forced oscillation

In a steady state, current and voltage oscillate at the frequency of the voltage source, which in this case is also called the excitation frequency.

If the excitation frequency corresponds to the resonance frequency, **resonance** occurs. This can be achieved by varying the excitation frequency or the component sizes L or C.

5.2 Definition of resonance condition, frequency, quality factor, characteristic impedance

For resonant circuits, when resonance occurs, the imaginary part of the impedance becomes zero. Respectively, the imaginary part of the admittance becomes zero and the current and voltage at the output terminals are in phase.

The resonance condition at the output terminals of the resonance circuit is:

$$\begin{aligned} \varphi_{u0} - \varphi_{i0} &\overset{!}{=} 0 \quad \text{Phase shift equal to zero} \\ \leftrightarrow \operatorname{Im}\{\underline{Z}_0\} &\overset{!}{=} 0 \quad \text{Reactance equal to zero} \end{aligned} \tag{5.1}$$

Frequency-variable quantities are marked with index 0 in the following resonance case:

$$\underline{Z}, \underline{U}, \underline{I}, \varphi_u, \varphi_i = \underline{Z}_0, \underline{U}_0, \underline{I}_0, \varphi_{u0}, \varphi_{i0} \quad \text{für} \quad \omega = \omega_0$$

The **resonance frequency** f_0 is the frequency at which excitation produces resonance or satisfies the resonance condition. In some cases, the **resonant circuit frequency** ω_0 is also imprecisely referred to as the resonance frequency. In general, $\omega_0 = 2\pi f_0$.

In simple RLC oscillating circuits, the reactance X_L of the inductance and X_C of the capacitance cancel each other out in the case of resonance. In terms of magnitude, these are equal, which is why their magnitude in the case of resonance is also referred to as the **characteristic impedance** X_k. Similarly, the **characteristic conductance** B_k is defined as the magnitude of the conductance of the inductance and the capacitance at resonance:

$$X_k = |X_{L,0}| = |X_{C,0}| \tag{5.2}$$

$$B_k = |B_{L,0}| = |B_{C,0}| = \frac{1}{X_k} \tag{5.3}$$

Characteristic resistance and conductance have the same units as impedance [Ω] and admittance [S].

The **quality factor** Q, also known as the resonance sharpness, is a dimensionless quantity. It is generally defined as the ratio of the reactive power of the inductance or capacitance to the (active) power of the oscillating circuit in the case of resonance:

$$Q = \frac{|Q_{L,0}|}{P_{R,0}} = \frac{|Q_{C,0}|}{P_{R,0}} \qquad \text{generally} \tag{5.4}$$

The quality factor is sometimes also defined specifically for series or parallel oscillating circuits.

For RLC series oscillating circuits, the quality factor corresponds to the ratio of characteristic impedance X_k to ohmic resistance R, while for RLC parallel oscillating circuits, the quality factor corresponds to the ratio of characteristic conductance B_k to ohmic conductance G:

$$Q = \frac{X_k}{R} \qquad \text{in series oscillating circuit} \tag{5.5}$$

$$Q = \frac{Y_k}{G} \qquad \text{in the parallel oscillating circuit} \tag{5.6}$$

The quality factor thus also describes the ratio of possible overvoltages to source voltages for series resonant circuits in the case of voltage resonance, or the ratio of overcurrents to source currents for parallel resonant circuits in the case of current resonance:

$$Q = \frac{U_{L,0}}{U_q} = \frac{U_{C,0}}{U_q} \qquad \text{in voltage resonance} \tag{5.7}$$

$$Q = \frac{I_{L,0}}{I_q} = \frac{I_{C,0}}{I_q} \qquad \text{in electrical resonance} \tag{5.8}$$

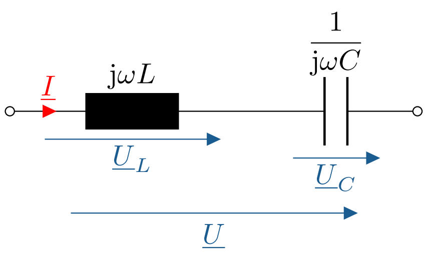
5.3 Resonant frequency using the example of ideal LC oscillating circuits

In example **??**, the resonance circuit frequency for ideal LC series and LC parallel oscillating circuits is determined.

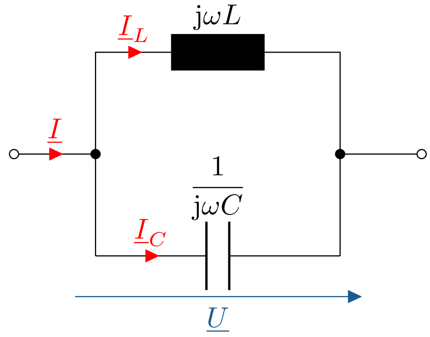
The examples use calculations to illustrate why resonance is caused by voltages in series resonant circuits and by currents in parallel resonant circuits. To distinguish between the two, we therefore refer to **voltage-related** and **current-related** resonance.

Example 5.1: Resonance in LC resonant circuits

Berechnung der Resonanzfrequenz ω_0 für LC-Serien- und LC-Parallelschwingkreise.



LC series resonant circuit



LC parallel oscillating circuit

When energized with voltage $\underline{U}$:

$$\begin{aligned}\underline{U} &= \underline{I}\cdot\left(\mathrm{j}\omega L + \frac{1}{\mathrm{j}\omega C}\right)\\ &= \underline{I}\cdot\mathrm{j}\left(\underbrace{\omega L - \frac{1}{\omega C}}_{=0\ \text{Resonanz}}\right)\end{aligned}$$

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\Rightarrow \underline{I}\to\infty \quad \text{für} \quad \underline{U} = konst.$$

Voltage-related resonance
in the series oscillating circuit

When energized by electricity $\underline{I}$:

$$\begin{aligned}\underline{I} &= \underline{U}\cdot\left(\mathrm{j}\omega C + \frac{1}{\mathrm{j}\omega L}\right)\\ &= \underline{U}\cdot\mathrm{j}\left(\underbrace{\omega C - \frac{1}{\omega L}}_{=0\ \text{Resonanz}}\right)\end{aligned}$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$

$$\Rightarrow \underline{U}\to\infty \quad \text{für} \quad \underline{I} = konst.$$

Current-related resonance
in the parallel oscillating circuit

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The **resonance frequency** is the same for both oscillating circuits.

In the case of resonance, the voltages across L and C cancel each other out in an ideal series resonant circuit. Therefore, theoretically, arbitrarily high voltages can arise across L and C. In real circuits, however, the voltage is limited by the ohmic resistance R. [See RLC resonant circuit in section 5.4]

The same applies in the case of resonance for the currents through L and C in a parallel resonant circuit.

The resonance frequency of simple RLC resonant circuits is identical to that of ideal LC resonant circuits, since the resistance R does not affect the reactive component of the impedance, with R, L and C all in series or all in parallel.

5.4 RLC series resonant circuit - resonance behaviour

This subchapter examines the resonance behaviour of an RLC series resonant circuit as an example. The resonant circuit is shown in Figure 5.4 and corresponds to the ideal LC series resonant circuit from Example **??** with an additional ohmic resistance R in series.

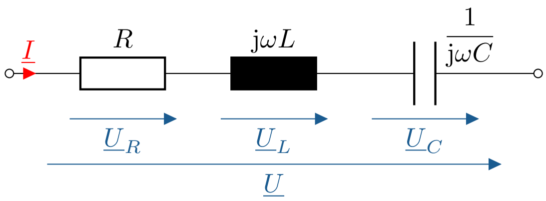


Figure 5.4: Circuit diagram of an RLC series resonant circuit

The impedance $\underline{Z}$ of the RLC series resonant circuit shown in Figure 5.4 is given by:

$$\underline{U} = \underline{I} \cdot \left(R + \mathrm{j}\omega L - \mathrm{j}\frac{1}{\omega C} \right) \tag{5.9}$$

$$\underline{Z} = R + \mathrm{j}\left(\omega L - \frac{1}{\omega C} \right) \tag{5.10}$$

In the case of resonance, according to equation 5.1, the imaginary part of the impedance disappears. Consequently, in this arrangement, the reactances of inductance and capacitance cancel each other out. By inserting $\underline{Z}$ into the resonance condition, the resonance frequency ω_0 can be determined:

$$\begin{aligned} \mathrm{Im}\{\underline{Z}\} = \left(\omega L - \frac{1}{\omega C} \right) &\stackrel{!}{=} 0 \\ \omega_0 L - \frac{1}{\omega_0 C} = 0 \quad &\Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}} \end{aligned} \tag{5.11}$$

Since the reactance of the inductance is positive and proportional to the frequency ($X_L \sim \omega$), and since the reactance of the capacitance is negative and proportional to the reciprocal of the frequency ($-X_C \sim \frac{1}{\omega}$), the inductive component predominates for frequencies above the resonance frequency and the capacitive component for frequencies below it.

This results in the following case distinction for the RLC series resonant circuit:

$$\mathrm{Im}\{\underline{Z}\} = \begin{cases} \omega L - \frac{1}{\omega C} < 0 & \omega < \omega_0 & \text{ohmic-capacitive} \\ \omega L - \frac{1}{\omega C} = 0 & \omega = \omega_0 & \text{purely ohmich} \\ \omega L - \frac{1}{\omega C} > 0 & \omega > \omega_0 & \text{ohmic-inductive} \end{cases} \tag{5.12}$$

The case differentiation is illustrated in Figure 5.5 using a pointer diagram and in Figure 5.6 using an impedance curve.

Digression: Free oscillation as a limiting case of resonance

The term resonance, from the Latin *resonare* (to echo), always refers, in accordance with the meaning of the word, to system states in the event of external excitation.

As shown in Figure 5.1, a charged RLC series resonant circuit oscillates at its natural frequency in a state of free oscillation when the external terminals are short-circuited.

The short circuit can be regarded as an ideal voltage source (internal resistance $R_i = 0$) with voltage $U = 0$. The natural frequency (with damping) is obtained from the solution of the homogeneous differential equation of the oscillating circuit. More on this in Module 12: Switching Processes.

Knot and mesh equations:

$$u(t) = u_R + u_L + u_C \overset{!}{=} 0$$
$$i(t) = i_R = i_L = i_C$$

Differential equation:

$$u(t) = iR + L\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{1}{C}\int i\mathrm{d}t \overset{!}{=} 0$$
$$L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{1}{C}i \overset{!}{=} 0$$

Approach:

$$i(t) = k\mathrm{e}^{\lambda t} \qquad \frac{\mathrm{d}i}{\mathrm{d}t} = k\lambda\mathrm{e}^{\lambda t} \qquad \frac{\mathrm{d}^2 i}{\mathrm{d}t^2} = k\lambda^2\mathrm{e}^{\lambda t}$$

Insertion:

$$L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{1}{C}i = 0$$
$$k \cdot \Big(\underbrace{L\lambda^2 + R\lambda + \frac{1}{C}}_{=0}\Big)\mathrm{e}^{\mathrm{j}\lambda t} = 0$$

Dissolve:

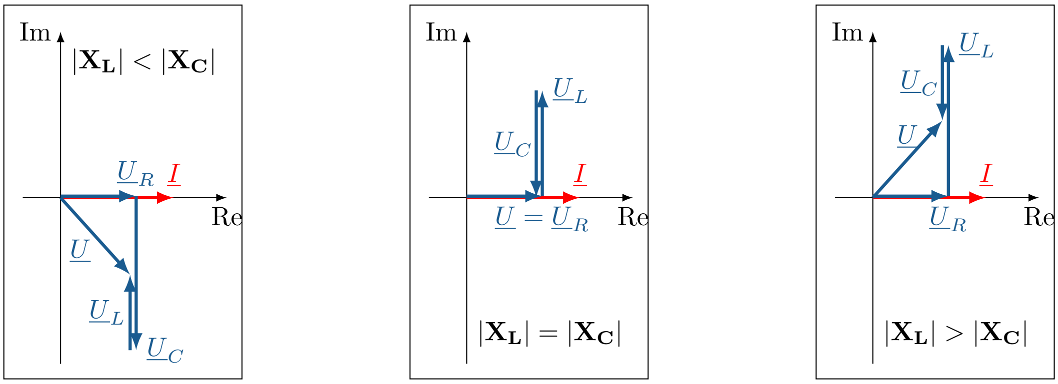
$$\lambda_{1/2} = -\frac{R}{2L} \pm \mathrm{j}\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

As explained in more detail in Module 12, if $\frac{1}{LC} > \left(\frac{R}{2L}\right)^2$ (low damping), a complex solution results. Only in this case can the resonant circuit oscillate freely, whereby the real part of the solution defines the decay constant and the imaginary part (root term) defines the natural frequency ω_d:

$$\omega_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

5.4.1 Case differentiation in the pointer diagram

Figure 5.5 shows a pointer diagram for all three cases. The voltages across R, L and C are shown, as well as the total voltage $\underline{U}$ when excited with a constant current $\underline{I}$. The true-to-scale length ratios $\frac{U_L}{U_C}$ correspond to: $\frac{1}{2}$, $\frac{\sqrt{2}}{\sqrt{2}}$, $\frac{2}{1}$.



(a) Fall 1: $\omega < \omega_0$ (ohmic-capacitive)

(b) Fall 2: $\omega = \omega_0$ (purely ohmic)

(c) Fall 3: $\omega > \omega_0$ (ohmic-inductive)

Figure 5.5: Pointer diagram of an RLC series resonant circuit

Since this is a passive resonant circuit (load arrow system), the active power P of the resonant circuit is always positive with $P = \mathrm{Re}\{\underline{U} \cdot \underline{I}\}$.

With a real impressed current $\underline{I}$, the real part of the total voltage $\underline{U}$ is therefore always positive. The pointer of the total voltage $\underline{U}$ therefore points either to the first quadrant (resistive-inductive), to the fourth quadrant (resistive-capacitive) or to the real axis (purely resistive).

5.4.2 Case differentiation in impedance curve

For the series resonant circuit shown in Figure 5.4, the impedance is given by:

$$\begin{aligned} \underline{Z} &= R + \mathrm{j}\,(X_L + X_C) \\ &= R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right) \end{aligned} \tag{5.13}$$

Capacitive reactance (hyperbolic) predominates for frequencies below the resonance frequency, while inductive reactance (linear) predominates for frequencies above the resonance frequency as shown in equation 5.12.

Figure 5.6 shows the impedance curve of an RLC series resonant circuit as a function of frequency relative to the resonance frequency f/f_0 equivalent to ω/ω_0. Shown are the magnitudes of the impedance $|\underline{Z}|$, the reactance $|X|$, the reactance of the inductance $|X_L|$, the reactance of the capacitance $|X_C|$ and the ohmic resistance R.

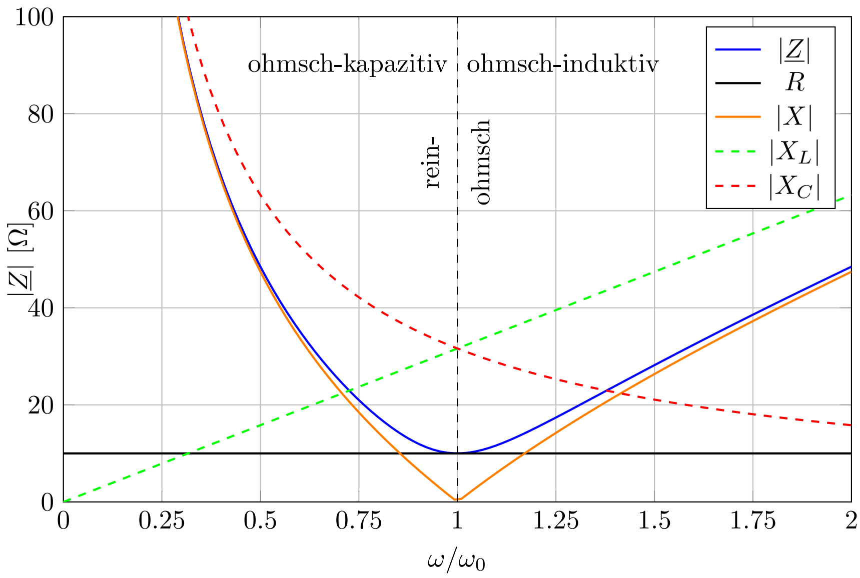


Figure 5.6: Impedance curve of an RLC series resonant circuit
mit $R = 10\ \Omega$, $L = 10$ mH, $C = 10\ \mu$F

As can be seen in the impedance curve, the impedance in the resonance case is purely ohmic and corresponds to the ohmic resistance R:

$$\underline{Z}_0 = R \tag{5.14}$$

The reactance of the inductance X_L (positive) and the reactance of the capacitance X_C (negative) cancel each other out in the case of resonance (intersection of the curves). The magnitudes of both reactances are equal in the case of resonance and correspond to the characteristic impedance X_k:

$$\begin{aligned} X_k = |X_{L,0}| = |X_{C,0}| &= \sqrt{\frac{L}{C}} \\ = \omega_0 L \quad &= \frac{1}{\omega_0 C} \end{aligned} \tag{5.15}$$

It can also be seen that the impedance value approaches the capacitive curve for very low frequencies $\omega << \omega_0$ with $\lim_{\omega\to 0} |Z| = |X_C|$ and for very high frequencies $\omega >> \omega_0$ approaches the inductive curve with $\lim_{\omega\to\infty} |Z| = |X_L|$.

5.4.3 Resonance curve and quality factor of an RLC series resonant circuit

The impedance of an RLC series resonant circuit can be expressed, as in equation 5.10, in terms of the component values R, L and C as a function of frequency ω:

$$\underline{Z} = R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right) \tag{5.16}$$

Using the resonant circuit frequency from equation 5.11, the angular frequency can be normalised to this value. To do this, the ω of the reactance terms are extended with ω_0 ($\omega = \omega\frac{\omega_0}{\omega_0}$). This allows the reactances X_L and X_C to be expressed using the characteristic impedance X_k from Eq. 5.15:

$$\begin{aligned}\underline{Z} &= R + \mathrm{j}\left(\omega L \frac{\omega_0}{\omega_0} - \frac{1}{\omega C}\frac{\omega_0}{\omega_0}\right)\\ &= R + \mathrm{j}\left(X_{L,0}\frac{\omega}{\omega_0} + X_{C,0}\frac{\omega}{\omega_0}\right)\\ &= R + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k\end{aligned} \tag{5.17}$$

The following applies:

$$X_L = +X_k \frac{\omega}{\omega_0} \qquad = +\sqrt{\frac{L}{C}} \cdot \frac{\omega\sqrt{LC}}{1} = \omega L \qquad \text{(linear)} \tag{5.18}$$

$$X_C = -X_k \frac{\omega_0}{\omega} \qquad = -\sqrt{\frac{L}{C}} \cdot \frac{1}{\omega\sqrt{LC}} = -\frac{1}{\omega C} \qquad \text{(hyperbolically)} \tag{5.19}$$

$$X_k = \sqrt{\frac{L}{C}} \qquad \text{mit} \quad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \text{(constant)} \tag{5.20}$$

As stated in equation 5.5, the quality factor Q of a series resonant circuit is the ratio of its characteristic resistance X_k to its ohmic resistance R. This allows the impedance of the resonant circuit to be expressed in terms of its characteristic resistance X_k using the quality factor Q:

$$\frac{\underline{Z}}{X_k} = \frac{1}{Q} + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \qquad \text{mit} \quad Q = \frac{X_k}{R} \tag{5.21}$$

The term $\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}$ is also referred to as relative detuning ν_r.[4]

Figure 5.7 shows several resonance curves of an RLC series resonant circuit for different quality factors Q for comparison. The y-axis shows the impedance normalised to the characteristic resistance X_k and the x-axis shows the frequency normalised to the resonance frequency ω_0.

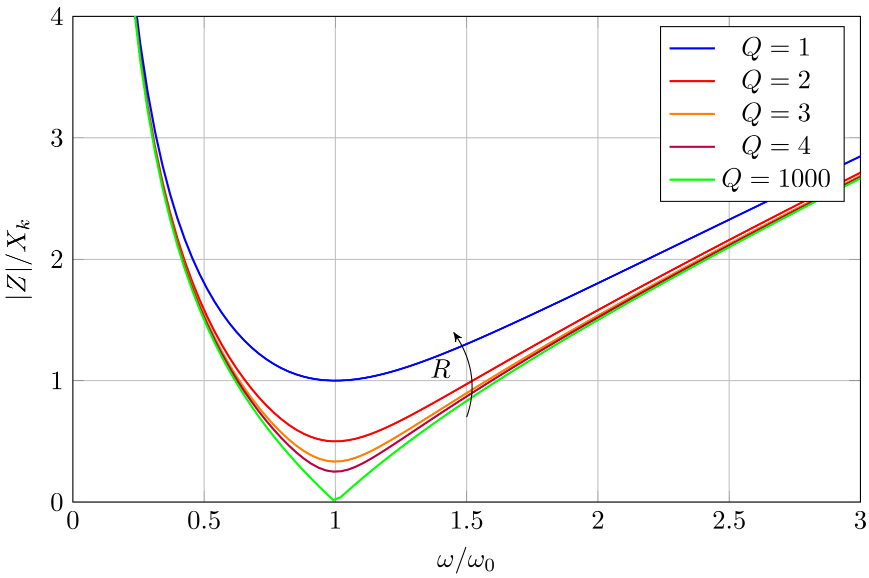


Figure 5.7: Impedance resonance curve, RLC series resonant circuit, comparison of quality factors

The most significant difference between the individual impedance curves for different quality factors can be seen in the resonance frequency range. As shown in equation 5.14, the following applies in the case of resonance: $Z_0 = R$.

A high quality factor means that the resonant circuit has a very low impedance relative to the characteristic resistance X_k at the resonance frequency. The quality factor is therefore also a measure of the resonant circuit's ability to oscillate or, conversely, of its damping.

The following applies to the series resonant circuit:

$$Q = \frac{X_k}{R} = \frac{\sqrt{\frac{L}{C}}}{R} \tag{5.22}$$

For frequencies well below the cut-off frequency $\omega << \omega_0$ and well above $\omega >> \omega_0$, the reactive components of the impedance dominate, as shown in Chapter 5.4. Therefore, the influence of the quality factor on the impedance at these frequencies is small (approximation of the curves).

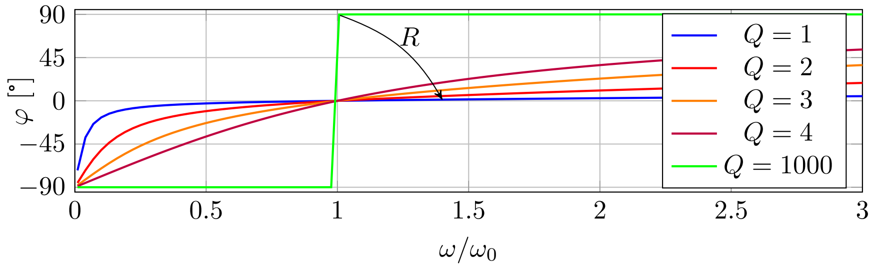


Figure 5.8: Resonance curve of the phase, RLC series resonant circuit, comparison of quality factors

Figure 5.8 shows the phase curve of the resonant circuit for different qualities with $\varphi = \varphi_U - \varphi_I$. For frequencies significantly below the resonance frequency $\omega << \omega_0$, the phase shift φ is almost $-90°$ (purely capacitive). For frequencies significantly above the resonance frequency $\omega >> \omega_0$, the phase shift φ is almost $+90°$ (purely inductive).

The higher the quality factor (resonance sharpness), the steeper the transition from capacitive to inductive in the resonance frequency range.

5.4.4 Voltage resonance using the example of an RLC series resonant circuit

When a series resonant circuit is excited by a constant voltage source $\underline{U}_q$, a **voltage resonance** may occur, characterised by overvoltages at L and C.

Consider the RLC series resonant circuit from Figure 5.4 with impedance $\underline{Z}$ from Equation 5.10. The voltage $\underline{U}$ across the resonant circuit corresponds to the constant source voltage $\underline{U}_q$ at its external terminals as shown in Figure 5.9.

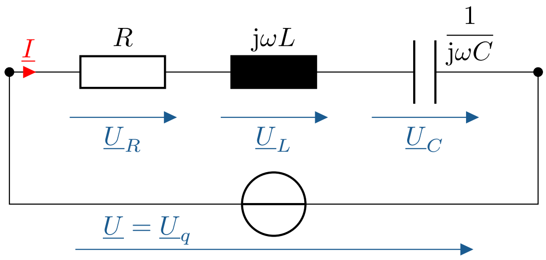


Figure 5.9: Circuit diagram of an RLC series resonant circuit with a constant voltage source

With the knot and mesh equation:

$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = \underline{U}_q$$
$$\underline{I} = \underline{I}_R = \underline{I}_L = \underline{I}_C$$

and the component impedances follow the complex voltage dividers:

$$\frac{\underline{U}_R}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{R}{R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)}$$
$$\frac{\underline{U}_L}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{\mathrm{j}\omega L}{R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)}$$
$$\frac{\underline{U}_C}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{\frac{1}{\mathrm{j}\omega C}}{R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)}$$

After shortening the currents, the frequency-variable impedance terms remain. The individual voltages are obtained by multiplying them by the voltage $\underline{U}$.

To normalise the frequency-dependent voltages, the reactances can be expressed as in equation 5.15 using the characteristic impedance X_k and the resonance circuit frequency ω_0 from equation 5.11. With the impedance normalised in this way from equation 5.17, the terms for the voltage dividers are obtained:

$$\frac{\underline{U}_R}{\underline{U}} = \frac{R}{R + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k}$$
$$\frac{\underline{U}_L}{\underline{U}} = \frac{\mathrm{j}X_k \frac{\omega}{\omega_0}}{R + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k}$$
$$\frac{\underline{U}_C}{\underline{U}} = \frac{-\mathrm{j}X_k \frac{\omega_0}{\omega}}{R + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k}$$

Figure 5.10 shows the magnitudes of the voltages U_R, U_L, U_C of the RLC series resonant circuit as a function of the angular frequency normalised to the resonant angular frequency. Normalisation to ω_0 allows, for example, the comparison of the quality of series resonant circuits with different resonant frequencies.

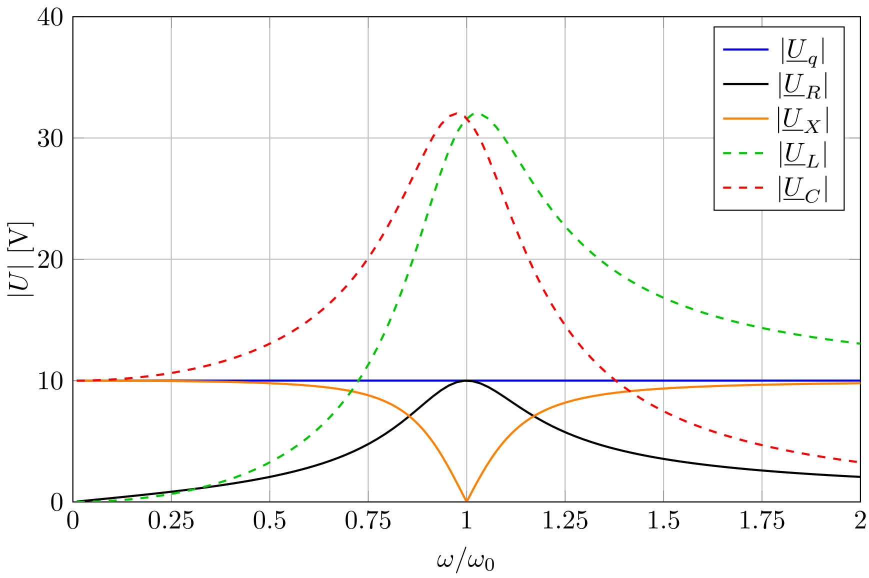


Figure 5.10: Resonance curve of the voltage at the RLC series resonant circuit
$U_q = 10$ V, $Q = 3{,}162$

As can be seen in the figure, in the resonance frequency range $\omega = \omega_0$, here is a significantly higher voltage across L and C compared to the ohmic resistance R.

The degree of voltage increase at the resonant frequency is expressed by the quality factor. As defined in equation 5.7, it corresponds to the ratio of the overvoltage of the apparent resistances (X_L, X_C) $U_{L,0}$, $U_{C,0}$ to the voltage of the effective resistance (R) $U_{R,0}$ in the case of resonance ($omega = \omega_0$).

$$Q = \frac{U_{L,0}}{U} = \frac{U_{C,0}}{U} = \frac{X_k}{R} \tag{5.23}$$

As can be seen in the graph, the voltage values $|U_L|$ and $|U_C|$ do not reach their maximum at the resonance frequency!

Digression: Derivation of quality via reactive power

In general, the following applies to the apparent power of a component:

$$\begin{aligned} \underline{S} &= S \cdot \mathrm{e}^{\mathrm{j}\varphi} && (5.24)\\ &= U \cdot I \cos\varphi + \mathrm{j}U \cdot I \sin\varphi && (5.25)\\ &= P \cdot \mathrm{j}Q && (5.26) \end{aligned}$$

As apparent resistances, inductance and capacitance can only absorb and emit reactive power. With $\varphi_L = +90°$ and $\varphi_C = -90°$, the following applies to the resonance case for the reactive power of both components:

$$\begin{aligned} Q_L &= U_L I_L \cdot \sin(+\frac{\pi}{2}) = +U_L \cdot I_L \\ Q_C &= U_C I_C \cdot \sin(-\frac{\pi}{2}) = -U_L \cdot I_L \end{aligned} \tag{5.27}$$

When excited by a constant voltage source $\underline{U}_q$, the same current $\underline{I}$ flows through all components in the series resonant circuit.

Using the general definition of quality via the reactive power in equation 5.4 of the inductance Q_L or the capacitance Q_C and the active power P of the resonant circuit in the case of resonance, the following applies to the series resonant circuit:

$$\begin{aligned} Q &= \frac{|Q_{L,0}|}{P_0} = \frac{U_{L,0} I_L}{U_{R,0} I_R} = \frac{U_{L,0} I}{U_{R,0} I} = \frac{U_{L,0}}{U} \\ &= \frac{|Q_{C,0}|}{P_0} = \frac{U_{C,0} I_C}{U_{R,0} I_R} = \frac{U_{C,0} I}{U_{R,0} I} = \frac{U_{C,0}}{U} \end{aligned} \tag{5.28}$$

As shown in Equation 5.28 and stated in Equation 5.7, the quality factor therefore corresponds to the ratio of the overvoltage of the apparent resistances (L, C) to the voltage of the effective resistance (R) in the resonance case.

Using the voltage divider rule, the quality factor can also be expressed as the ratio of the characteristic impedance to the effective impedance by substituting the reactances for the resonance case, as stated in equation 5.5. With X_k from Eq. 5.15 and $\underline{Z}$ from Eq. 5.10, the following applies:

$$\begin{aligned} \frac{U_{L,0}}{U} &= \frac{\cancel{I}}{\cancel{I}} \cdot \frac{|\mathrm{j} X_{L,0}|}{|R + \mathrm{j}(X_{L,0} + X_{C,0})|} \\ &= \frac{|X_{L,0}|}{|R + \mathrm{j}\underbrace{(X_{L,0} + X_{C,0})}_{=\,0}|} \\ &= \frac{|X_{L,0}|}{R} = \frac{X_k}{R} = Q \end{aligned} \tag{5.29}$$

5.4.5 Current curve of the RLC series resonant circuit

When excited with a constant voltage $\underline{U}_q$, the RLC series resonant circuit shown in Figure 5.9 results in a frequency-dependent current curve. This is shown in terms of magnitude in Figure 5.11 for different quality factors Q.

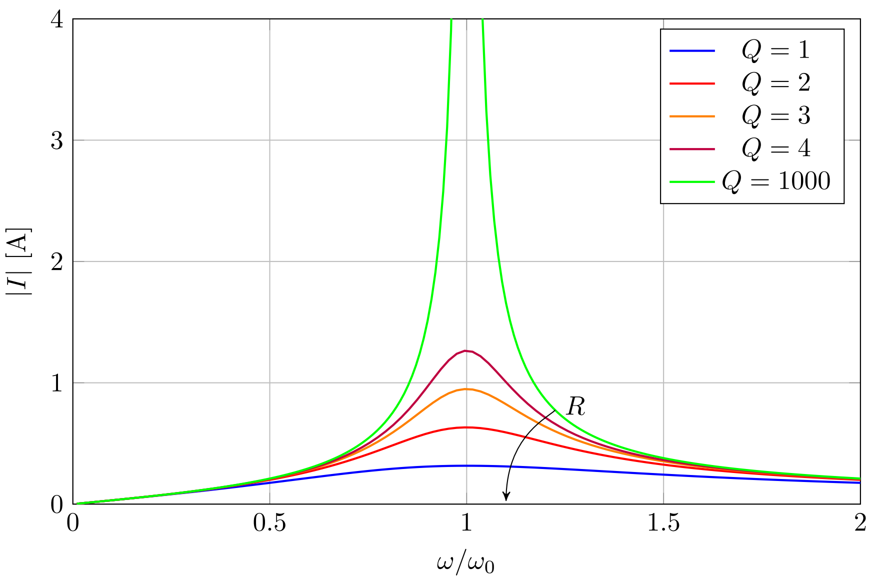


Figure 5.11: Current curve of the RLC series resonant circuit at constant voltage $X_k = 31{,}6\ \Omega$, $U_q = 10$ V, $Q = var.$

The current is at its maximum at the resonance frequency, as can be seen in the graph. As shown in Figure 5.6, the impedance is minimal in the case of resonance and corresponds to the ohmic resistance R.

In general, the following relationship applies to the current $\underline{I}$ with the impedance $\underline{Z}$ from equation 5.17 and the resonance frequency ω_0 from equation 5.11:

$$\begin{aligned} \underline{I} &= \frac{\underline{U}}{\underline{Z}} \\ &= \frac{\underline{U}_q}{R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)} \\ &= \frac{\underline{U}_q}{R + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k} \end{aligned} \tag{5.30}$$

The maximum current I_0 is obtained for a real source voltage $\underline{U}_q = U_q$ from the quotient of the source voltage divided by the resistance R:

$$I_0 = \frac{U_q}{R} \qquad \text{mit} \quad \underline{Z}_0 = R \tag{5.31}$$

This means that both the total current $\underline{I}$ and its active component $\mathrm{Re}\{\underline{I}\}$ are limited by the resistance R.

For the current $\underline{I}$ relative to the maximum current I_0, the quality factor from equation 5.28 applies:

$$\begin{aligned} \frac{\underline{I}}{I_0} &= \frac{U_q}{\underline{Z}} \cdot \frac{R}{U_q} = \frac{R}{\underline{Z}} \\ &= \frac{1}{1 + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q} \end{aligned} \tag{5.32}$$

5.4.6 Voltage curve of the RLC series resonant circuit

If an RLC series resonant circuit is excited with a constant current $\underline{I}_q$ as shown in Figure 5.12, component voltages result.

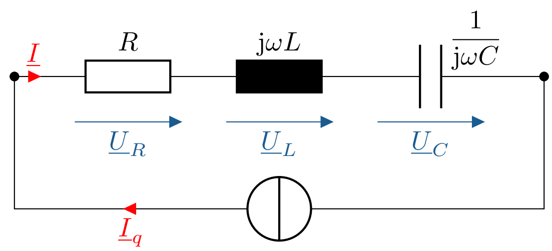


Figure 5.12: Circuit diagram of an RLC series resonant circuit with a constant current source

The component voltages are directly proportional to the respective impedances. The following applies to the total voltage:

$$\underline{U} = \underline{I}_q \cdot \underline{Z}$$

The voltage curve therefore corresponds exactly to the impedance curve in Figure 5.7.

5.5 RLC-Parallelschwingkreis - Resonanzverhalten

This subchapter examines the resonance behaviour of an RLC parallel resonant circuit. The resonant circuit is shown in Figure 5.13 and corresponds to the ideal LC parallel resonant circuit from Example **??** with an additional parallel ohmic resistance R.

The investigation is analogous to that for the RLC series resonant circuit in Chapter 5.4. In the parallel resonant circuit, the admittance $\underline{Y}$ is primarily considered. This results in similar case distinctions and calculation methods as in the series resonant circuit, whereby the roles of current and voltage, inductance and capacitance, as well as resistance and conductance are reversed:

$$\underline{Y} = \underline{Z}^{-1} = G + \mathrm{j}B = \frac{\underline{I}}{\underline{U}} \tag{5.33}$$

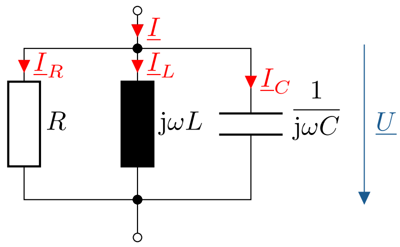


Figure 5.13: Circuit diagram of an RLC parallel resonant circuit

The admittance $\underline{Y}$ of the RLC parallel resonant circuit shown in Figure 5.13 is given by:

$$\underline{I} = \underline{U} \cdot (G + \mathrm{j}B_C + \mathrm{j}B_L) \tag{5.34}$$

$$\underline{Y} = G + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right) \tag{5.35}$$

In the case of resonance, according to equation 5.1, the imaginary part of the impedance or admittance disappears. Consequently, in this arrangement, the susceptances of inductance B_L and capacitance B_C cancel each other out. By inserting $\underline{Y}$ into the resonance condition, the resonance frequency ω_0 can be determined:

$$\begin{aligned} \mathrm{Im}\{\underline{Y}\} = \left(\omega C - \frac{1}{\omega L}\right) &\stackrel{!}{=} 0 \\ \omega_0 C - \frac{1}{\omega_0 L} = 0 \quad &\Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}} \end{aligned} \tag{5.36}$$

Since the susceptance of the capacitance is positive and proportional to the frequency ($B_C = \omega C \sim \omega$),

and since the susceptance of the inductance is negative and proportional to the reciprocal of the frequency ($-B_L \sim \frac{1}{\omega}$), the inductive component predominates for frequencies below the resonance frequency and the capacitive component predominates for frequencies above the resonance frequency. This results in the following case distinction for the RLC series resonant circuit:

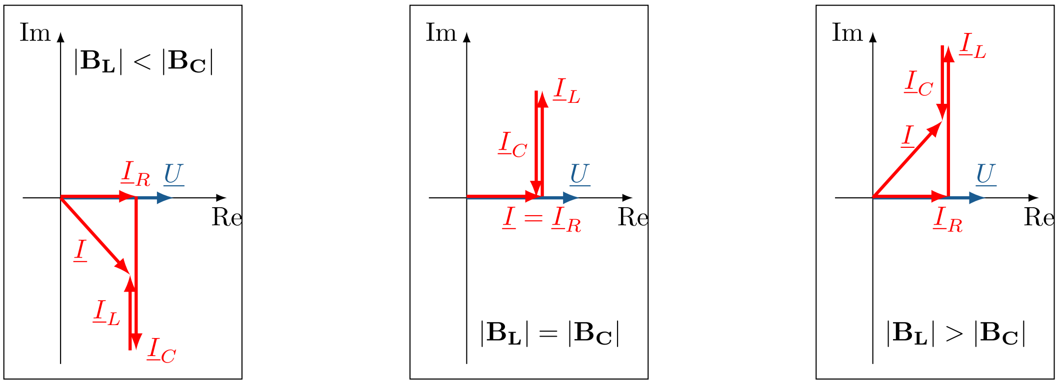
$$\mathrm{Im}\{\underline{Y}\} = \begin{cases} \omega C - \frac{1}{\omega L} < 0 & \omega < \omega_0 & \text{ohmsch-induktiv} \\ \omega C - \frac{1}{\omega L} = 0 & \omega = \omega_0 & \text{rein ohmsch} \\ \omega C - \frac{1}{\omega L} > 0 & \omega > \omega_0 & \text{ohmsch-kapazitiv} \end{cases} \tag{5.37}$$

The case differentiation is illustrated in Figure 5.14 using a pointer diagram and in Figure 5.15 using an admittance curve.

5.5.1 Case differentiation in the pointer diagram

Figure 5.14 shows an example of a phasor diagram for all three cases.

The diagram shows the currents flowing through R, L and C, as well as the total current $\underline{I}$ when excited with a constant voltage $\underline{U}$. The true-to-scale length ratios $\frac{I_L}{I_C}$ correspond to: $\frac{1}{2}$, $\frac{\sqrt{2}}{\sqrt{2}}$, $\frac{2}{1}$.



(a) Case 1: $\omega < \omega_0$ (ohmic-inductive)

(b) Case 2: $\omega = \omega_0$ (purely ohmic)

(c) Case 3: $\omega > \omega_0$ (ohmic-capacitive)

Figure 5.14: Pointer diagram of RLC parallel resonant circuit

Since this is a passive resonant circuit (consumer counting arrow system), the active power P of the resonant circuit is always positive with $P = \mathrm{Re}\{\underline{U} \cdot \underline{I}\}$.

When voltage $\underline{U}$ is applied (plotted with phase angle 0), the real component of the total current $\underline{I}$ is always positive. The pointer of the total current $\underline{I}$ therefore points either to the first quadrant (resistive-inductive), to the real axis (purely resistive) or to the fourth quadrant (resistive-capacitive).

5.5.2 Case differentiation in admittance curve

For the parallel oscillating circuit shown in Figure 5.13, the admittance is given by Equation 5.35:

$$\begin{aligned} \underline{Y} &= G + \mathrm{j}\,(B_C + B_L) \\ &= G + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right) \end{aligned} \tag{5.38}$$

The inductive susceptibility (hyperbolic) predominates for frequencies below the resonance frequency, while the capacitive susceptibility (linear) predominates for frequencies above the resonance frequency as shown in equation 5.37.

Figure 5.15 shows the admittance curve of an RLC parallel resonant circuit as a function of frequency relative to the resonance frequency ${}^{f}/_{f_0}$ equivalent to ${}^{\omega}/_{\omega_0}$. Shown are the magnitudes of the admittance $|\underline{Y}|$, the susceptance $|B|$, the susceptance of the capacitance $|B_C|$, the susceptance of the inductance $|B_L|$ and the ohmic conductance G.

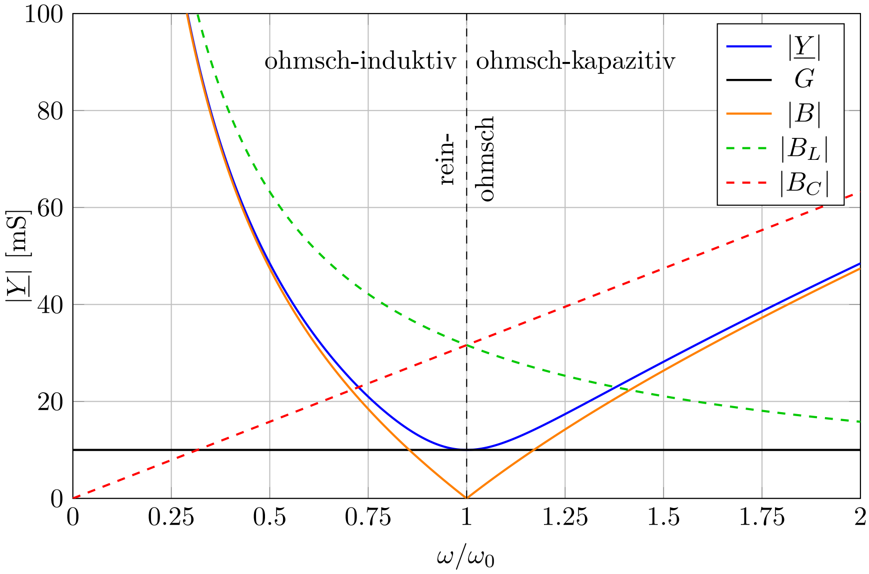


Figure 5.15: Admittance curve of an RLC parallel resonant circuit
with $R = 10\ \Omega$, $L = 10$ mH, $C = 10\ \mu$F

As can be seen in the admittance curve, the admittance is purely ohmic in the case of resonance and corresponds exactly to the ohmic conductance G:

$$\underline{Y}_0 = G \tag{5.39}$$

The susceptances of the capacitance B_C (positive) and the inductance B_L (negative) cancel each other out in the case of resonance (intersection of the curves). The magnitudes of both susceptances are equal in the case of resonance and correspond to the characteristic value B_k:

$$\begin{aligned} B_k = |B_{C,0}| = |B_{L,0}| &= \sqrt{\frac{C}{L}} \\ = \omega_0 C \quad &= \frac{1}{\omega_0 L} \end{aligned} \tag{5.40}$$

It can also be seen that the admittance approaches the inductive curve for very low frequencies $\omega << \omega_0$ with $\lim_{\omega\to 0} |Y| = |B_L|$ and for very high frequencies $\omega >> \omega_0$ approaches the capacitive curve with $\lim_{\omega\to\infty} |Y| = |B_C|$.

5.5.3 Resonance curve and quality factor of an RLC parallel resonant circuit

As in equation 5.35, the admittance of an RLC parallel resonant circuit can be expressed in terms of the component values R, L and C as a function of frequency ω as follows:

$$\underline{Y} = G + \mathrm{j}B = \frac{1}{R} + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right) \tag{5.41}$$

Using the resonant circuit frequency from equation 5.36, the circular frequency can be normalised to this. The calculation is performed analogously to the frequency normalisation for the impedance of the RLC series resonant circuit in equation 5.17. The susceptances B_C and B_L are described using the characteristic value B_k from Eq. 5.40 and the frequency ratio of ω to ω_0. The following applies to the admittance:

$$\begin{aligned}\underline{Y} &= G + \mathrm{j}\left(\omega C \frac{\omega_0}{\omega_0} - \frac{1}{\omega L}\frac{\omega_0}{\omega_0}\right)\\ &= G + \mathrm{j}\left(B_{C,0}\frac{\omega}{\omega_0} - B_{L,0}\frac{\omega}{\omega_0}\right)\\ &= G + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k\end{aligned} \tag{5.42}$$

The following applies:

$$B_C = +B_k \frac{\omega}{\omega_0} \qquad = \omega C \qquad \text{(linear)} \tag{5.43}$$

$$B_L = -B_k \frac{\omega_0}{\omega} \qquad = -\frac{1}{\omega L} \qquad \text{(hyperbolisch)} \tag{5.44}$$

$$B_k = \sqrt{\frac{C}{L}} \qquad \text{mit} \quad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \text{(konstant)} \tag{5.45}$$

As stated in equation 5.6, the quality Q of a parallel resonant circuit can be described by the ratio of the characteristic value B_k to the effective value G. In this respect, the definition differs from that in a series resonant circuit.

The general definition of quality in equation 5.4 based on the ratio of reactive power to active power can be used to derive the specific definition of quality for the parallel resonant circuit. The same applies to the series resonant circuit, as shown in equation 5.28:

$$\begin{aligned}Q &= \frac{|Q_{L,0}|}{P_0} = \frac{U \cdot I_{L,0}}{U \cdot I_R} = \frac{I_{L,0}}{I_R} = \frac{|B_{L,0}|}{G}\\ &= \frac{|Q_{C,0}|}{P_0} = \frac{U \cdot I_{C,0}}{U \cdot I_R} = \frac{I_{C,0}}{I_R} = \frac{|B_{C,0}|}{G} = \frac{B_k}{G}\end{aligned} \tag{5.46}$$

This allows the impedance of the oscillating circuit to be expressed in terms of its characteristic value B_k using the quality factor Q:

$$\frac{\underline{Y}}{B_k} = \frac{1}{Q} + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \qquad \text{mit} \quad Q = \frac{B_k}{G} \tag{5.47}$$

The term $\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}$ is also referred to as relative detuning ν_r. [4]

Figure 5.16 shows several resonance curves of an RLC parallel resonant circuit for different quality factors Q for comparison. The y-axis shows the admittance normalised to the characteristic value B_k and the x-axis shows the angular frequency normalised to the resonance frequency ω_0.

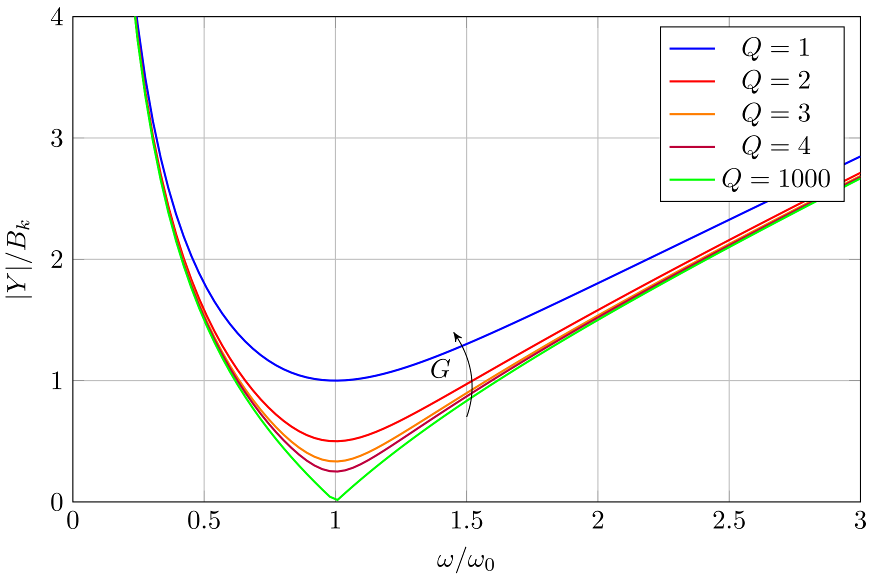


Figure 5.16: Resonanzkurve RLC-Parallelschwingkreis, Vergleich Gütefaktoren

The component sizes are the same as those chosen for the impedance curve of the RLC series resonant circuit in Figure 5.6. Equations 5.42 and 5.47 show that the normalised representation of both curves allows a direct comparison of the quality factors.

The most significant difference between the individual admittance curves for different quality factors can be seen in the resonance frequency range. With $\underline{Y}_0 = G = \frac{B_k}{Q}$, the admittance in the resonance case corresponds directly to the ohmic conductance G and is inversely proportional to the quality factor.

The quality factor is a measure of the oscillation capacity of the oscillating circuit and a measure of the inverse damping of the oscillating circuit. In contrast to a series resonant circuit, a high quality factor in a parallel resonant circuit means a low effective conductance G (high effective resistance R) in relation to a high reactive conductance B (low reactive resistance X).

5.5.4 Current resonance using the example of an RLC parallel resonant circuit

When a parallel resonant circuit is excited by a constant current source $\underline{I}_q$, a **current resonance** may occur, characterised by current flow between L and C.

Consider the RLC parallel resonant circuit shown in Figure 5.13 with the admittance $\underline{Y}$ from Equation 5.35. The current $\underline{I}$ through the resonant circuit corresponds to the constant source current $\underline{I}_q$ at its external terminals as shown in Figure 5.17.

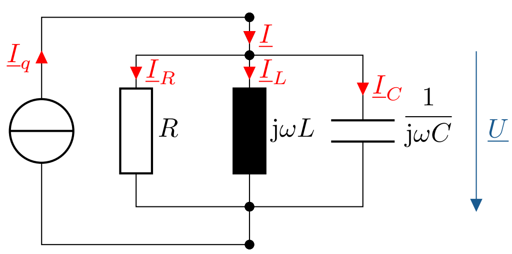


Figure 5.17: Circuit diagram of an RLC parallel resonant circuit with a constant current source

With the knot and mesh equation:

$$
\begin{aligned}
\underline{U} &= \underline{U}_R = \underline{U}_L = \underline{U}_C \\
\underline{I} &= \underline{I}_R + \underline{I}_L + \underline{I}_C = \underline{I}_q
\end{aligned}
$$

and the component admittances are followed by the complex current dividers:

$$
\begin{aligned}
\frac{\underline{I}_R}{\underline{I}} &= \frac{\underline{U}}{\underline{U}} \cdot \frac{\underline{Y}_R}{\underline{Y}} = \frac{G}{G + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right)} \\
\frac{\underline{I}_L}{\underline{I}} &= \frac{\underline{U}}{\underline{U}} \cdot \frac{\underline{Y}_L}{\underline{Y}} = \frac{\frac{1}{\mathrm{j}\omega L}}{G + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right)} \\
\frac{\underline{I}_C}{\underline{I}} &= \frac{\underline{U}}{\underline{U}} \cdot \frac{\underline{Y}_C}{\underline{Y}} = \frac{\mathrm{j}\omega C}{G + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right)}
\end{aligned}
$$

The current ratios of component currents to total current correspond to the frequency-variable ratio of component admittance to total admittance. The calculation of the current divider via the admittances is suitable in parallel circuits due to the simple addition of the individual admittances, similar to the calculation via the impedance in series circuits.

To normalise the frequency-dependent currents, the susceptances can be expressed as in Equation 5.40 using the characteristic value B_k and the resonance circuit frequency ω_0 from Equation 5.36. With the admittance normalised in this way from equation 5.42, the following terms result for the voltage dividers:

$$
\begin{aligned}
\frac{\underline{I}_R}{\underline{I}} &= \frac{G}{G + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k} \\
\frac{\underline{I}_L}{\underline{I}} &= \frac{-\mathrm{j}\frac{\omega_0}{\omega} B_k}{G + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k} \\
\frac{\underline{I}_C}{\underline{I}} &= \frac{\mathrm{j}\frac{\omega}{\omega_0} B_k}{G + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k}
\end{aligned}
\tag{5.48}
$$

Figure 5.18 shows the magnitudes of the currents I_R, I_L, I_C as a function of the angular frequency normalised to the resonance angular frequency.

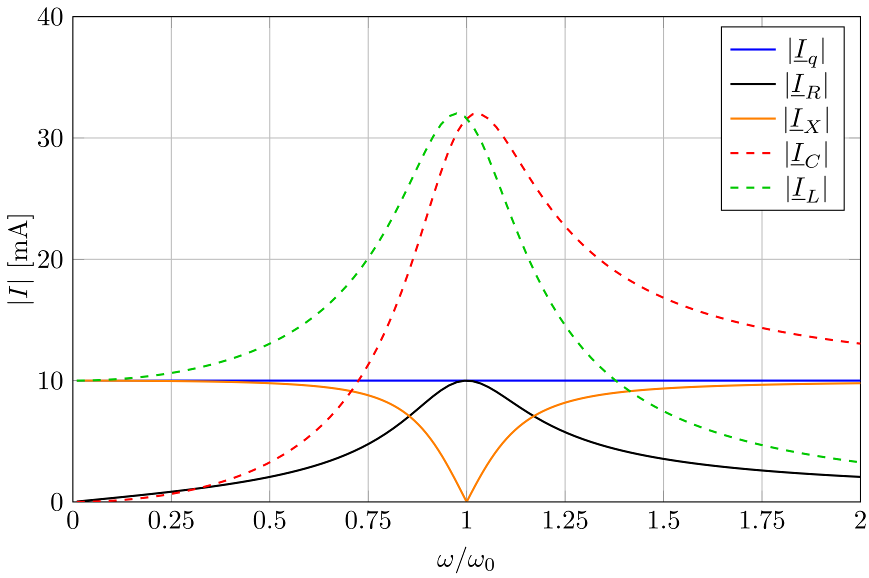


Figure 5.18: Resonance curve of the current at the RLC parallel resonant circuit
$U_q = 10$ V, $Q = 3{,}162$

As can be seen in the figure, in the range of the resonance circuit frequency $\omega = \omega_0$, there is a current increase at L and C compared to the ohmic resistance R. The respective maximum current is not exactly at ω_0, but slightly below the resonance frequency for L and slightly above it for C.

As shown in equation 5.46 and stated in equation 5.8, the quality corresponds to the ratio of the overcurrent to the total current in the resonance case. In the case of resonance, $I_0 = I_{R,0} = I_{q,0}$. Therefore, the quality of the current divider rule can be expressed as the ratio of the characteristic conductance B_k to the effective conductance G:

$$Q = \frac{I_{L,0}}{I} = \frac{I_{C,0}}{I} = \frac{B_k}{G} \tag{5.49}$$

The frequency-dependent current conditions expressed by the quality factor Q with $\underline{Y}$ from equation 5.47:

$$\begin{aligned}
\frac{\underline{I}_R}{\underline{I}} &= \frac{\frac{1}{Q}}{\frac{1}{Q} + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)} \\
\frac{\underline{I}_L}{\underline{I}} &= \frac{-\mathrm{j}\frac{\omega_0}{\omega}}{\frac{1}{Q} + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)} \\
\frac{\underline{I}_C}{\underline{I}} &= \frac{\mathrm{j}\frac{\omega}{\omega_0}}{\frac{1}{Q} + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}
\end{aligned} \tag{5.50}$$

5.5.5 Voltage curve of the RLC parallel resonant circuit

When excited with a constant current $\underline{I}_q$ as shown in Figure 5.17, the RLC parallel resonant circuit from Figure 5.17. This is shown in terms of magnitude in Figure 5.19 for different quality factors Q.

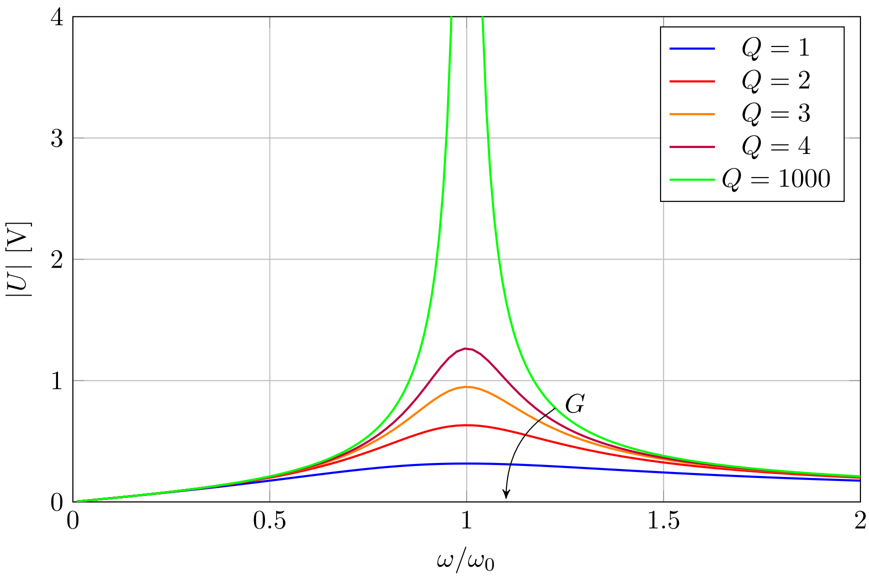


Figure 5.19: Spannungskurve des RLC-Parallelschwingkreises bei konstantem Strom $B_k = 31{,}6$ mS, $I_q = 10$ mA, $Q = var.$

The voltage behaviour of the RLC parallel resonant circuit is analogous to the current behaviour of the RLC series resonant circuit, as described in Chapter 5.4.5.

With the admittance normalised to the characteristic value from equation 5.47, Kirchhoff's rules and Ohm's law, the following applies to the voltage $\underline{U}$:

$$\begin{aligned} \underline{U} &= \frac{\underline{I}}{\underline{Y}} \\ &= \frac{\underline{I}_q}{G + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)} \\ &= \frac{\underline{I}_q}{G + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k} \end{aligned} \tag{5.51}$$

The maximum voltage U_0 occurs in the case of resonance. It corresponds to the quotient of the real source current $\underline{I}_q = I_q$ by the effective conductance G:

$$U_0 = \frac{I_q}{G} \qquad \text{mit} \quad \underline{Y}_0 = G \tag{5.52}$$

This limits both the complex total voltage $\underline{U}$ and its effective component $\mathrm{Re}\{\underline{U}\}$ by the conductance G and the source current I_q.

For the voltage $\underline{U}$ normalised to the maximum voltage U_0, the following results from Eq. 5.51:

$$\begin{aligned} \frac{\underline{U}}{U_0} &= \frac{\frac{\underline{I}_q}{\underline{Y}}}{\frac{I_q}{G}} = \frac{G}{\underline{Y}} \\ &= \frac{1}{1 + \mathrm{j}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q} \end{aligned} \tag{5.53}$$

The normalised voltage curve of the RLC parallel resonant circuit therefore corresponds exactly to the normalised current curve of the RLC series resonant circuit with Figure 5.10 and Equation 5.32.

5.5.6 Current curve of the RLC parallel resonant circuit

The current curve of the RLC parallel resonant circuit when excited by a constant voltage source $\underline{U}_q$ is derived from the admittance $\underline{Y}$ and the voltage source $\underline{U}_q$:

$$\underline{I} = \underline{U}_q \cdot \underline{Y}$$

This means that the current $\underline{I}$ is directly proportional to the admittance $\underline{Y}$ and thus to the angular frequency ω. The current curve of the RLC parallel resonant circuit therefore corresponds to the admittance curve of the resonant circuit as shown in Figure 5.16.

6 Additional calculation

Frequency response of a first-order RC low-pass filter. [Additional calculation for equation (2.6)]

$$\begin{aligned}
\underline{F}(\mathrm{j}\omega) &= \frac{\underline{U}_2}{\underline{U}_1} && \Big| = \frac{Z_C}{(Z_R + Z_C)} \frac{\cdot \cancel{\underline{I}}}{\cdot \cancel{\underline{I}}} && \text{Complex voltage divider rule} \\
&= \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}} && \Big| \cdot \frac{\mathrm{j}\omega C}{\mathrm{j}\omega C} && \text{Rationalise} \\
&= \frac{1}{\mathrm{j}\omega CR + 1} && \Big| && \text{Transform} \\
&= \frac{1}{1 + \mathrm{j}\omega CR} && \Big| \cdot \frac{1 - \mathrm{j}\omega CR}{1 - \mathrm{j}\omega CR} && \text{Optional for Cartesian form} \\
&= \frac{1 - \mathrm{j}\omega CR}{1 + (\omega CR)^2}
\end{aligned} \tag{6.1}$$

A Exercise tasks

A.1 RC low-pass filter, 1st order

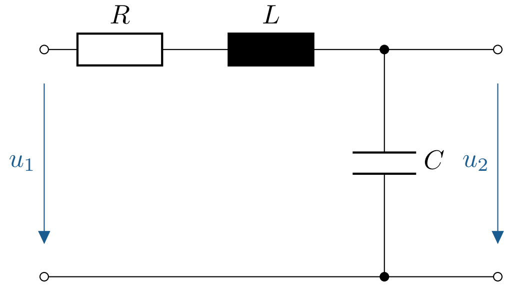
The output voltage of an RC low-pass filter should only be 10What should the cut-off frequency be?

A.2 1st order RC high pass filter

What is the phase shift φ in a first-order RC high-pass filter if the stopband attenuation (ratio of output voltage to input voltage) is $-6dB$?

A.3 Frequency response of an RLC series resonant circuit

Given is the circuit shown on the right with $L = 20\,\text{mH}$, a resistor $R = 150\,\Omega$ and a capacitor with $C =$ 1μF.



a) Specify the resonance frequency of the circuit. Is it current or voltage resonance?

b) Derive the frequency response $\frac{\underline{U}_2}{\underline{U}_1}$ in general terms.

c) Calculate and sketch the value of the frequency response $\frac{\underline{U}_2}{\underline{U}_1}$ in semi-logarithmic representation. Give the magnitude for $\omega = 0, 100\,\text{s}^{-1}$, $2000\,\text{s}^{-1}$, $3500\,\text{s}^{-1}$, ω_0 , $10000\,\text{s}^{-1}$ and ∞.

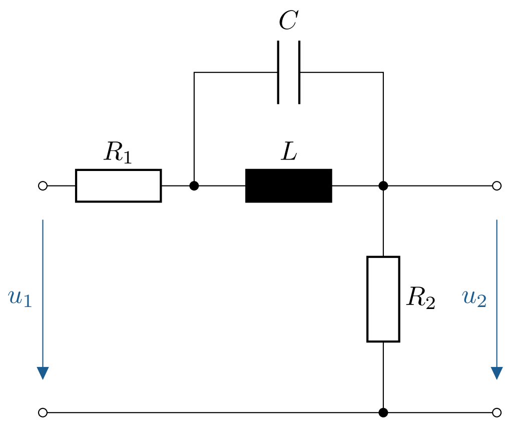
d) Does an inductance of 10 mH lead to voltage overvoltage in the event of resonance?

A.4 Frequency response of a four-pole circuit consisting of R, L and C

The circuit shown on the right is to be examined. The general values R_1, R_2, L, C are to be used for the components.

a) Determine the voltage divider ratio.

$$\underline{F}(\mathrm{j}\omega) = \frac{\underline{U}_2}{\underline{U}_1}$$



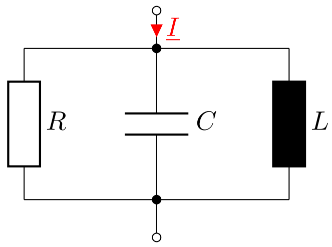
b) Calculate and sketch the magnitude and phase of $\underline{F}(\mathrm{j}\omega)$. Mark characteristic points.

c) Calculate the output voltage $u_2(t)$ if the input voltage is

$$u_1(t) = 5\text{V} \cdot \cos(2\pi \cdot 10\text{kHz} \cdot t)$$

gilt.

A.5 RLC parallel resonant circuit



The oscillating circuit shown on the right is to be examined.

a) Determine the complex total impedance $\underline{Z}$ in general form.

b) Sketch the amount of complex impedance in a linear representation and mark characteristic points.

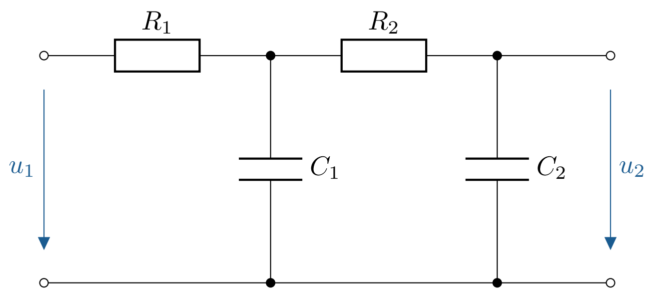
c) Calculate the resonance frequency f_0 as well as the frequencies f_u and f_o at which the magnitude of $\underline{Z}(\omega)$ has dropped to $\frac{Z_0}{\sqrt{2}}$, where Z_0 is the magnitude of the total impedance at resonance. Also determine the difference between the two frequencies. This difference represents the bandwidth B. For this part of the problem, assume $R = 300\,\Omega$, $L = 200\,\mathrm{mH}$, and $C = 1\,\mu\mathrm{F}$.

d) The bandwidth is also determined using the formula $B = \frac{f_0}{Q}$. Compare the result with that from part c) of the exercise.

e) Are the upper and lower cut-off frequencies symmetrical around f_0?

A.6 Second-order RC filter

The following applies to the RC elements connected in series below $R_1 = R_2 = R = 1000\,\Omega$ and $C_1 = C_2 = C = 1\,\mu\mathrm{F}$.



a) Determine the frequency response of the circuit. $\underline{F}(\mathrm{j}\omega) = \frac{\underline{U}_2}{\underline{U}_1}$.

b) Plot the amplitude response and phase response in logarithmic representation. Compare the result with the simple RC circuit with the same components.

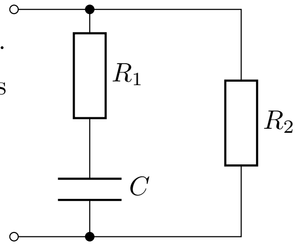
A.7 Impedance and admittance locus curve of an RL element

Given is the series connection of an inductance $L = 10\,\mathrm{mH}$ and a resistance $R = 5\,\mathrm{k\Omega}$ at a variable frequency with $f_\mathrm{min} = 5\,\mathrm{kHz}$ and $f_\mathrm{max} = 50\,\mathrm{kHz}$.

a) Determine the impedance location curve and represent it quantitatively. Mark the parameter at least three meaningful points.

b) Draw the inverted locus curve (admittance locus curve). First, consider the basic curve and then use some numerical values from the previous task a) to determine the locus curve points.

A.8 Admittance locus curve of mixed series/parallel connection



The admittance of the circuit shown on the right is to be investigated.

Parallel to a series connection of resistor R_1 and capacitor C, there is a second resistor. R_2.

The component values are given by:
$R_1(p) = p \cdot R_0$, $\quad R_0 = 5\,\Omega$, $\quad R_2 = 20\,\Omega \quad$ und $\quad \omega C = 5\,\Omega$

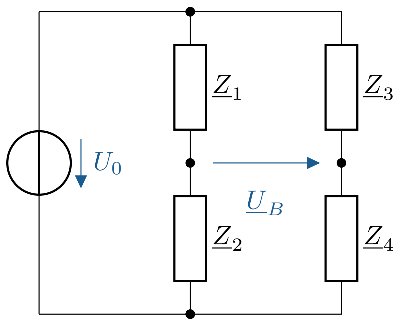
a) Determine the admittance for $p = 0$, $p = 1$, $p = 2$ and $p \to \infty$.

b) Sketch the course of the admittance curve and plot the values. (Recommended display range: $0 + \mathrm{j}0$ bis $0{,}25\,\mathrm{S} + \mathrm{j}0{,}25\,\mathrm{S}$)

c) For what value of p does the total current $\underline{I}$ of an applied alternating voltage $\underline{U}$ lead by 45°? Determine the value for $\underline{Y}$ by drawing a diagram.

d) Check the result from c) mathematically.

A.9 AC measuring bridge



a) Determine the general equation for the measured voltage $\underline{U}_B$. Bring the result to a common denominator and simplify if possible.

b) A reactive resistance is to be measured using the differential method. In the given measurement range, let $\underline{Z}_1 = \mathrm{j}X_1$ and $\underline{Z}_2 = \mathrm{j}X_2$ (pure reactive resistances) and $\underline{Z}_3 = \underline{Z}_4 = R$ (pure active resistances). Determine the equation for the measured voltage $\underline{U}_B$.

c) Is the measured voltage $\underline{U}_B$ from task b) frequency-dependent? Justify your answer.

d) How does the measurement voltage behave? $\underline{U}_B$ bei $X_1 = X_2$?

e) How does the measurement voltage behave? $\underline{U}_B$ bei $X_1 = -X_2$?

B Solutions to the exercises

B.1 RC low-pass filter, 1st order

$$\underline{F}(j\omega) = \frac{\underline{U}_a}{\underline{U}_e} = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} = \frac{1}{1 + j\omega RC}$$

$$A(\omega) = \frac{U_a}{U_e} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$A(\omega_g) = \frac{1}{\sqrt{1 + (\omega_g RC)^2}} \stackrel{!}{=} \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \omega_g = \frac{1}{RC}$$

$$A(\omega) \stackrel{!}{=} 0{,}1 \quad \Rightarrow \quad \sqrt{1 + (\omega RC)^2} \stackrel{!}{=} 10$$

$$\Leftrightarrow \quad 1 + (\omega RC)^2 = 100 \quad \Leftrightarrow \quad (\omega RC)^2 = 99 \quad \Rightarrow \quad \omega_g = \frac{\omega}{\sqrt{99}}$$

$$\Rightarrow \quad f_g = \frac{f}{\sqrt{99}} = \frac{1\,\text{kHz}}{\sqrt{99}} = 100{,}5\,\text{Hz}$$

B.2 1st order RC high pass filter

$$-6\text{ dB} = 20 \cdot \lg\left(\frac{U_a}{U_e}\right)\text{ dB}$$

$$\Rightarrow \quad -0{,}3 = \lg\left(\frac{U_a}{U_e}\right)$$

$$\Rightarrow \quad \frac{U_a}{U_e} = 10^{-0{,}3} = 0{,}501$$

$$\frac{\underline{U}_a}{\underline{U}_e} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{1}{1 - j\frac{1}{\omega CR}}$$

$$\frac{U_a}{U_e} = \frac{1}{\sqrt{1 + \frac{1}{(\omega CR)^2}}} \stackrel{!}{=} 0{,}501$$

$$\Leftrightarrow \quad \sqrt{1 + \frac{1}{(\omega CR)^2}} = \frac{1}{0{,}501}$$

$$\Leftrightarrow \quad \frac{1}{(\omega CR)^2} = 2{,}981 \quad \Leftrightarrow \quad \frac{1}{\omega CR} = \sqrt{2{,}981} = 1{,}727$$

$$\frac{\underline{U}_a}{\underline{U}_e} = \frac{1 + j\frac{1}{\omega CR}}{1 + \frac{1}{(\omega CR)^2}}$$

$$\varphi = \arctan\left(\frac{1}{\omega CR}\right) = \arctan\left(\sqrt{2{,}981}\right) = 59{,}92^\circ$$

B.3 Frequency response of an RLC series resonant circuit

a) The resonance frequency ω_0 is given by the resonance condition $\text{Im}\{\underline{Z} = 0\}$.

$$\underline{Z} = R + j\omega L + \frac{1}{j\omega C} = R + j \cdot \left(\omega L + \frac{1}{\omega C}\right) \qquad \Rightarrow \quad \omega_0 L - \frac{1}{\omega_0 C} \stackrel{!}{=} 0$$

$$\Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20\,\text{mH} \cdot 1\,\mu\text{F}}} = 7071{,}07\,\text{s}^{-1} \qquad \Rightarrow \quad f_0 = \frac{\omega_0}{2\pi} = 1125{,}4\,\text{Hz}$$

Since the circuit is a series connection of a resistor, capacitor and coil, it is a voltage resonance.

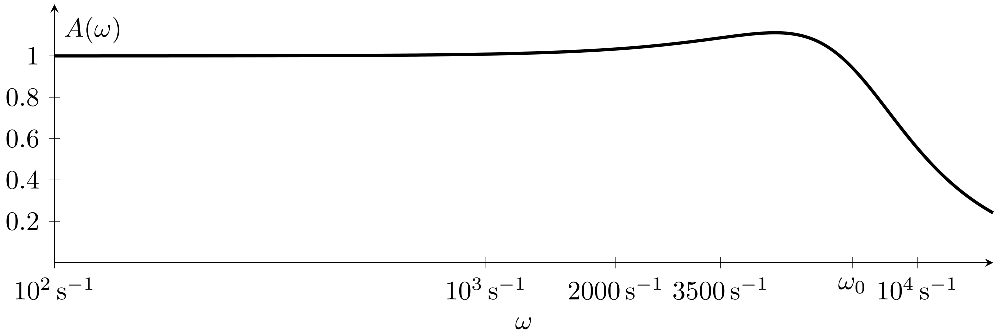
b) The voltage divider rule provides the frequency response:

$$\begin{aligned}\frac{\underline{U}_2}{\underline{U}_1} = \underline{F}(\mathrm{j}\omega) &= \frac{\underline{Z}_\mathrm{C}}{\underline{Z}_\mathrm{R}+\underline{Z}_\mathrm{L}+\underline{Z}_\mathrm{C}} \\ &= \frac{\frac{1}{\mathrm{j}\omega C}}{R+\mathrm{j}\omega L+\frac{1}{\mathrm{j}\omega C}} \qquad \Big| \cdot \frac{\mathrm{j}\omega C}{\mathrm{j}\omega C} \\ &= \frac{1}{\mathrm{j}\omega CR+\mathrm{j}^2\omega^2 LC+1} \\ &= \frac{1}{1-\omega^2 LC+\mathrm{j}\omega RC}\end{aligned}$$

c) The amount of the frequency response (amplitude response $A(\omega)$) is calculated as follows:

$$\left|\frac{\underline{U}_2}{\underline{U}_1}\right| = A(\omega) = \frac{1}{\sqrt{(1-\omega^2 LC)^2+(\omega RC)^2}}$$

The amount of frequency response is shown in the following figure.



The corresponding values are:

ω	0	$100\,\mathrm{s}^{-1}$	$2000\,\mathrm{s}^{-1}$	$3500\,\mathrm{s}^{-1}$	ω_0	$10000\,\mathrm{s}^{-1}$	$\omega\to\infty$
$A(\omega)$	1,000	1,009	1,033	1,087	0,943	0,555	0

d) The quality of a series resonant circuit indicates the voltage increase across L or C in the case of resonance.

$$\begin{aligned}\frac{U_2(\omega_0)}{U_1(\omega_0)} = Q_S = \frac{X_k}{R} = \frac{\omega_0 L}{R} &= \frac{1}{R}\cdot\sqrt{\frac{L}{C}} \\ &= \frac{1}{150\,\Omega}\cdot\sqrt{\frac{10\,\mathrm{mH}}{1\,\mu\mathrm{F}}} = 0{,}\overline{66}\end{aligned}$$

This $L = 10\,\mathrm{mH}$ ist $Q_S < 1$, means that there is no voltage increase in the event of resonance.

B.4 Frequency response of a four-pole circuit consisting of R, L and C

a) The voltage divider ratio is calculated as follows:

$$\begin{aligned}\underline{F}(\mathrm{j}\omega) &= \frac{\underline{U}_2}{\underline{U}_1} \\ &= \frac{R_2}{R_1+\underline{Z}_L||\underline{Z}_C+R_2} \qquad \text{mit} \quad \underline{Z}_L||\underline{Z}_C = \frac{\mathrm{j}\omega L\cdot\frac{1}{\mathrm{j}\omega C}}{\mathrm{j}\omega L-\mathrm{j}\frac{1}{\omega C}} = -\mathrm{j}\cdot\left(\frac{1}{\omega C-\frac{1}{\omega L}}\right) \\ &= \frac{R_2}{R_1+R_2-\mathrm{j}\left(\frac{1}{\omega C-\frac{1}{\omega L}}\right)}\end{aligned}$$

b) The voltage divider ratio is generally calculated as follows:

$$|\underline{F}(\mathrm{j}\omega)| = A(\omega) = \frac{\mathrm{R}_2}{\sqrt{(\mathrm{R}_1+\mathrm{R}_2)^2 + \left(\frac{1}{\omega\mathrm{C}-\frac{1}{\omega\mathrm{L}}}\right)^2}}$$

The parallel connection of L and C behaves like a short circuit for $\omega \to 0$ and for $\omega \to \infty$, since one of the two components has an impedance of zero in both cases. This results in the following for $\omega \to 0$ and $\omega \to \infty$:

$$A(\omega \to 0) = A(\omega \to \infty) = \frac{R_2}{R_1 + R_2}$$

In the case of resonance ($\omega = \omega_0 = \sqrt{\frac{1}{LC}}$), the admittances of L and C (parallel resonant circuit) cancel each other out. With $Y_C + Y_L \to 0$, it follows that $Z_L||Z_C \to \infty$ for $\omega \to \omega_0$.

$$A(\omega_0) = \frac{R_2}{\sqrt{(\mathrm{R}_1+\mathrm{R}_2)^2 + \infty^2}} = \frac{1}{\infty} = 0$$

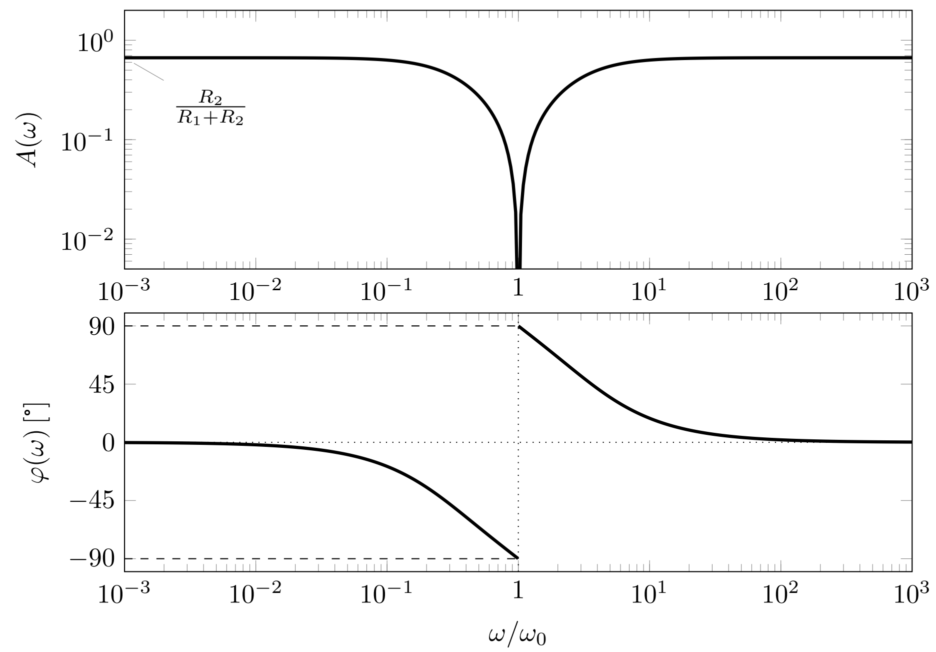
The phase results in:

$$\varphi(\omega) = \underbrace{\arctan\left(\frac{0}{R_2}\right)}_{\varphi_{\text{Zähler}}} - \underbrace{\arctan\left(\frac{-\left(\frac{1}{\omega C-\frac{1}{\omega L}}\right)}{R_1+R_2}\right)}_{\varphi_{\text{Nenner}}} = \arctan\left(\frac{\frac{1}{\omega C-\frac{1}{\omega L}}}{R_1+R_2}\right)$$

With:

$$\begin{aligned}
\varphi(\omega \to 0) &= \arctan(0) = 0 \\
\varphi(\omega \to \infty) &= \arctan(0) = 0 \\
\varphi(\omega_0) &= \arctan(\infty) = \pm\frac{\pi}{2} \\
\varphi(\omega < \omega_0) &< 0 \\
\varphi(\omega > \omega_0) &> 0
\end{aligned}$$

The amplitude response and phase response are shown in the following Bode diagram.



c) The output voltage is calculated as follows:

$$\begin{aligned}\underline{U}_2 &= \underline{F}(\mathrm{j}\omega)\cdot\underline{U}_1\\ u_2(t) &= A(\omega)\cdot 5\,\mathrm{V}\cdot\cos\left(\omega t+\varphi(\omega)\right) \qquad \text{mit} \quad \omega = 2\pi\cdot 10\,\mathrm{kHz}\cdot\end{aligned}$$

B.5 RLC parallel resonant circuit

a) The total impedance of the resonant circuit is calculated as follows:

$$\begin{aligned}\underline{Y} &= \frac{1}{R}+\mathrm{j}\omega C+\frac{1}{\mathrm{j}\omega L}\\ \underline{Z} &= \frac{1}{\underline{Y}} = \frac{1}{\frac{1}{R}+\mathrm{j}\omega C+\frac{1}{\mathrm{j}\omega L}}\\ &= \frac{\frac{1}{R}-\mathrm{j}\cdot\left(\omega C-\frac{1}{\omega L}\right)}{\frac{1}{R^2}+\left(\omega C-\frac{1}{\omega L}\right)^2}\end{aligned}$$

b) Amount of total impedance in semi-logarithmic representation:

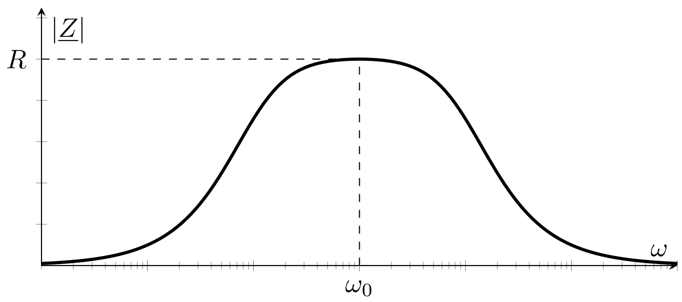


Figure B.1

The total impedance is at its maximum at the resonance frequency with $\underline{Z}(\omega_0) = R$.

c) Resonance:

$$\begin{aligned}\mathrm{Im}\{\underline{Y}\} &= \omega_0 C-\frac{1}{\omega_0 L} \overset{!}{=} 0 \quad \Leftrightarrow \omega_0 = \frac{1}{\sqrt{LC}} \quad \Rightarrow f_0 = \frac{\omega_0}{2\pi}\\ |\underline{Z}(\omega_0)| &= \frac{1}{\sqrt{\frac{1}{R^2}+\underbrace{\left(\omega_0 C-\frac{1}{\omega_0 L}\right)^2}_{=0}}} = \frac{1}{\sqrt{\frac{1}{R^2}}} = R\end{aligned}$$

Cutoff frequencies, bandwidth:

$$|\underline{Z}(\omega)| \stackrel{!}{=} \frac{R}{\sqrt{2}} \quad \Leftrightarrow \quad \frac{1}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} = \frac{R}{\sqrt{2}}$$

$$\frac{R}{\sqrt{1 + R^2 \cdot \left(\omega C - \frac{1}{\omega L}\right)^2}} = \frac{R}{\sqrt{2}}$$

$$1 + R^2 \cdot \left(\omega C - \frac{1}{\omega L}\right)^2 = 2$$

$$\left(\omega C - \frac{1}{\omega L}\right)^2 = \frac{1}{R^2}$$

$$\omega C - \frac{1}{\omega L} = \pm\frac{1}{R}$$

$$\omega^2 LC \pm \omega \cdot \frac{L}{R} - 1 = 0$$

$$\omega^2 \pm \omega \cdot \frac{1}{RC} - \frac{1}{LC} = 0$$

$$x^2 + p \cdot x + q = 0$$

$$x_{1,2} = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

$$\Rightarrow \omega = \pm\frac{1}{2 \cdot RC} \pm \sqrt{\frac{1}{(2 \cdot RC)^2} + \frac{1}{LC}}$$

$$\frac{1}{2 \cdot RC} = 1{,}6667 \cdot 10^3 \text{ s}^{-1} \qquad \frac{1}{LC} = 5 \cdot 10^6 \text{ s}^{-2}$$

$$\omega = \pm 1{,}6667 \cdot 10^3 \text{ s}^{-1} \pm 2{,}7889 \cdot 10^3 \text{ s}^{-1}$$

$$\omega g{,}o = 4{,}4556 \cdot 10^3 \text{s}^{-1} \qquad f_{g,o} = 709{,}12 \text{ Hz}$$

$$\omega g{,}u = 1{,}1222 \cdot 10^3 \text{s}^{-1} \qquad f_{g,u} = 178{,}60 \text{ Hz}$$

$$B = f_{g,o} - f_{g,u} = 530{,}5\,\text{Hz}$$

d) Determination of bandwidth via resonance frequency and quality factor:

$$B = \frac{f_0}{Q} \qquad \text{mit} \quad Q = \sqrt{\frac{C}{L}} \cdot R = \sqrt{\frac{1\ \mu\text{F}}{200\ \text{mH}}} \cdot 300\ \Omega = 0{,}6708$$

$$B = \frac{355{,}88\ \text{Hz}}{0{,}6708} = 530{,}5\ \text{Hz} \qquad \text{mit} \quad f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{200\ \text{mH} \cdot 1\ \mu\text{F}}} = 355{,}88\ \text{Hz}$$

e) With a logarithmic frequency axis, the curve of the impedance magnitude $|\underline{Z}(\omega)|$ is symmetrical around the resonance frequency ω_0, i.e. $f_{g,o}$ and $f_{g,u}$ are symmetrical around f_0.

$$\left(\omega C - \frac{1}{\omega L}\right)^2 = \bigg(\underbrace{\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}}_{=\vartheta(\omega),\ \vartheta^2 \text{ symmetrisch um } \omega_0 \text{ bei faktorieller Änderung}}\bigg)^2 \cdot \frac{C}{L} \qquad \Longrightarrow \qquad \vartheta^2(n \cdot \omega_0) = \vartheta^2\left(\frac{1}{n} \cdot \omega_0\right)$$

B.6 Second-order RC filter

a) Frequency response:

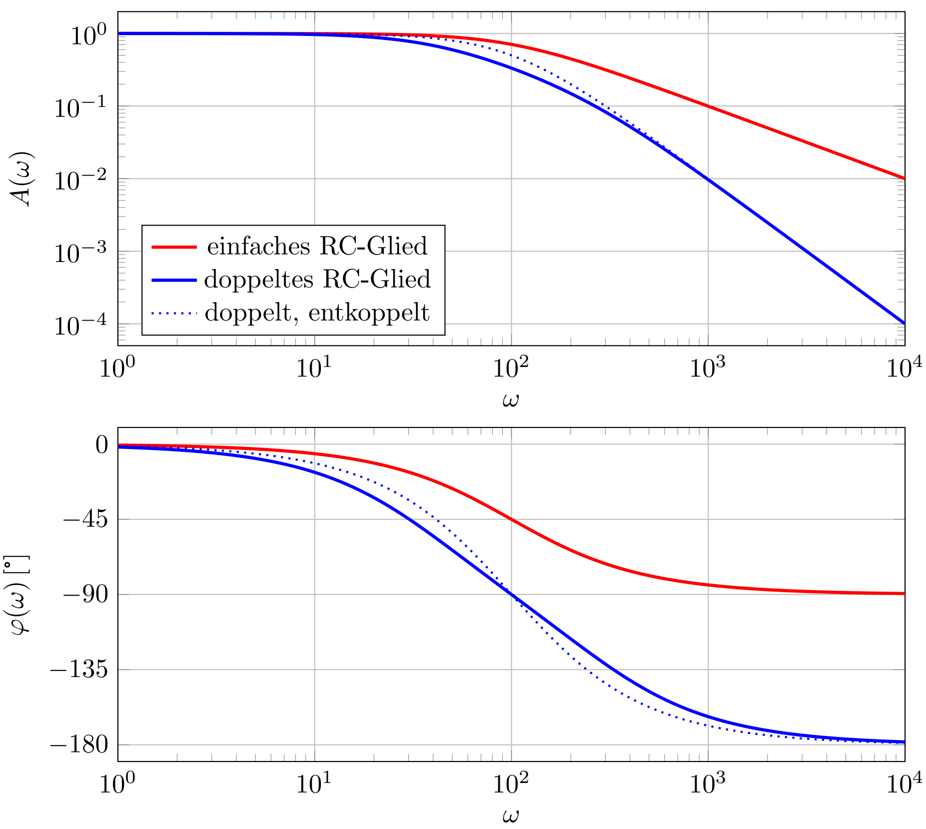
$$
\begin{aligned}
\underline{F}(\mathrm{j}\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{\underline{U}_2}{\underline{U}_m} \cdot \frac{\underline{U}_m}{\underline{U}_1} \\
\frac{\underline{U}_m}{\underline{U}_1} &= \frac{\underline{Z}_{\mathrm{ers}}}{R + \underline{Z}_{\mathrm{ers}}} = \frac{1}{\frac{R}{\underline{Z}_{\mathrm{ers}}} + 1} \\
\underline{Z}_{\mathrm{ers}} &= \left(R + \frac{1}{\mathrm{j}\omega C}\right) \Bigg|\Bigg| \frac{1}{\mathrm{j}\omega C} = \left(\frac{1}{R + \frac{1}{\mathrm{j}\omega C}} + \mathrm{j}\omega C\right)^{-1} \\
\frac{\underline{U}_m}{\underline{U}_1} &= \frac{1}{R \cdot \left(\frac{1}{R + \frac{1}{\mathrm{j}\omega C}} + \mathrm{j}\omega C\right) + 1} \\
&= \frac{1}{\frac{R \cdot \mathrm{j}\omega C}{R \cdot \mathrm{j}\omega C + 1} + R \cdot \mathrm{j}\omega C + 1} \\
&= \frac{R \cdot \mathrm{j}\omega C + 1}{R \cdot \mathrm{j}\omega C + (R \cdot \mathrm{j}\omega C + 1)^2} \\
&= \frac{R \cdot \mathrm{j}\omega C + 1}{R \cdot \mathrm{j}\omega C - R^2\omega^2 C^2 + 2 \cdot \mathrm{j}\omega C R + 1} \\
&= \frac{1}{1 - \omega^2 C^2 R^2 + \mathrm{j}3\omega C R} \\
&= \frac{1 + \mathrm{j}\Omega}{1 - \Omega^2 + \mathrm{j}3\Omega} \qquad \text{mit} \qquad \Omega = \omega C R \\
\frac{\underline{U}_2}{\underline{U}_m} &= \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}} = \frac{1}{1 + \mathrm{j}\omega C R} = \frac{1}{1 + \mathrm{j}\Omega} \\
\frac{\underline{U}_2}{\underline{U}_1} &= \frac{\underline{U}_2}{\underline{U}_m} \cdot \frac{\underline{U}_m}{\underline{U}_1} = \frac{1}{1 + \mathrm{j}\omega C R} = \frac{1}{1 + \mathrm{j}\Omega} \cdot \frac{1 + \mathrm{j}\Omega}{1 - \Omega^2 + \mathrm{j}3\Omega} \\
&= \frac{1}{1 - \Omega^2 + \mathrm{j}3\Omega} = \frac{1}{1 - (\omega C R)^2 + \mathrm{j}3\omega C R}
\end{aligned}
$$

Comparison of simple low-pass filter:

$$
\underline{F}(\mathrm{j}\omega) = \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}} = \frac{1}{1 + \mathrm{j}\omega C R} = \frac{1}{1 + \mathrm{j}\Omega}
$$

Due to coupling effects, the frequency response of the simple low-pass filter cannot simply be multiplied in series connection.

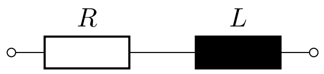
b) Bode diagram (amplitude and phase response) for 1st and 2nd order RC low-pass filters:



The attenuation in the blocking range is twice as high (40 dB /dec) for the second order (blue) compared to the first order (red) (20 dB /dec). The phase shift ranges from 0° to −180° for the second order, in contrast to the first order, which only ranges from 0° to −90°. The cut-off frequency shifts to the left (smaller for the second order) due to the higher attenuation.

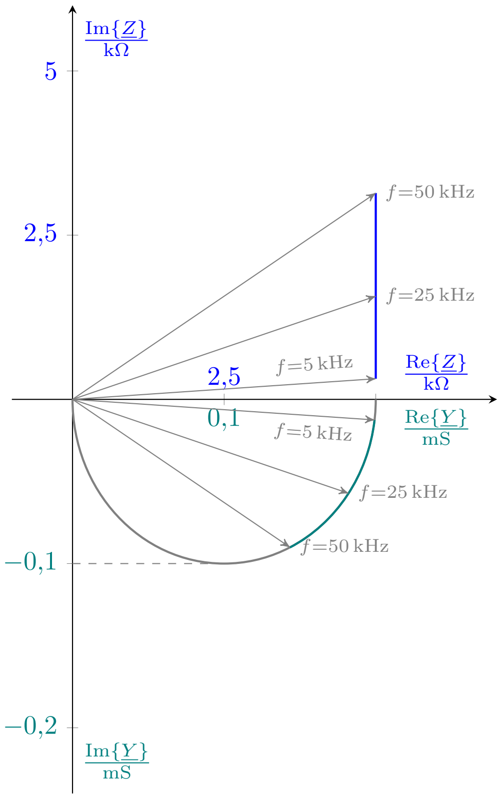
B.7 Impedance and admittance locus curve of an RL element

a)



$$f = 5\,\mathrm{kHz}...50\,\mathrm{kHz} \qquad \underline{Z} = R + \mathrm{j}\omega L$$
$$f_1 = 5\,\mathrm{kHz} \qquad \underline{Z}_1 = 5\,\mathrm{k\Omega} + \mathrm{j}314{,}16\,\Omega$$
$$f_2 = 25\,\mathrm{kHz} \qquad \underline{Z}_2 = 5\,\mathrm{k\Omega} + \mathrm{j}1570{,}8\,\Omega$$
$$f_3 = 50\,\mathrm{kHz} \qquad \underline{Z}_3 = 5\,\mathrm{k\Omega} + \mathrm{j}3141{,}6\,\Omega$$

$\underline{Z}$-Locus curve: $\mathrm{Re}\{\underline{Z}\} = konst.$, $\mathrm{Im}\{\underline{Z}\} = var.$
$\Rightarrow$ Exactly parallel to the imaginary axis in the first quadrant.



b) Inversion of straight lines in the first quadrant: Semicircle in the fourth quadrant.

$$\underline{Y} = \frac{1}{\underline{Z}} \qquad \underline{Y} = \frac{1}{|\underline{Z}|} \cdot \mathrm{e}^{-\mathrm{j}\varphi z}$$

The scale is chosen so that $\underline{Z}(0) = 5\,\mathrm{k\Omega} \hat{=} \underline{Y}(0) = 0{,}2\,\mathrm{mS}$.

Start of the semicircle at 0,2mS ($\omega = 0$), end at 0 mS ($\omega \to \infty$) and centre point at 0,1 mS.

The admittances can be read at the respective intersection of the mirrored impedance vector with the semicircle (angle negated, mirroring on the x-axis).

The admittance curve corresponds to the circular section between $\underline{Y}(f = 5\,\mathrm{kHz})$ und $\underline{Y}(f = 50\,\mathrm{kHz})$.

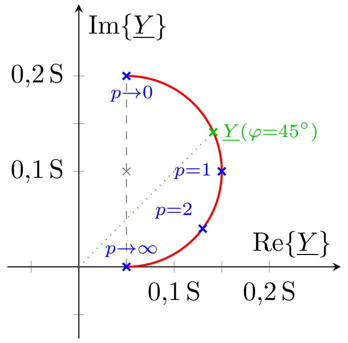
B.8 Admittance locus curve of mixed series/parallel connection

a) Admittance $\underline{Y}$ für $p = 0$, $p = 1$, $p = 2$ und $p \to \infty$:

$$\begin{aligned}
\underline{Y} &= \frac{1}{R_2} + \frac{1}{R_1 + \frac{1}{\mathrm{j}\omega C}} \\
&= \frac{1}{R_2} + \frac{R_1 - \frac{1}{\mathrm{j}\omega C}}{R_1^2 - \left(\frac{1}{\mathrm{j}\omega C}\right)^2} \\
&= \frac{1}{R_2} + \frac{p \cdot R_0 + \mathrm{j}\frac{1}{\omega C}}{p^2 \cdot R_0^2 + \left(\frac{1}{\omega C}\right)^2} \\
&= \frac{1}{20\,\Omega} + \frac{p \cdot 5\,\Omega + \mathrm{j} \cdot 5\,\Omega}{p^2 \cdot 25\,\Omega^2 + 25\,\Omega^2} \\
&= 0{,}05\,\mathrm{S} + \frac{p + \mathrm{j}}{p^2 + 1} \cdot 0{,}2\,\mathrm{S}
\end{aligned}$$

$$\begin{aligned}
\underline{Y}(p) &= \left(0{,}05 + \frac{p + \mathrm{j}}{p^2 + 1} \cdot 0{,}2\right)\,\mathrm{S} \\
\underline{Y}(p \to 0) &= \left(0{,}05 + \frac{0 + \mathrm{j}}{0 + 1} \cdot 0{,}2\right)\,\mathrm{S} = (0{,}05 + \mathrm{j} \cdot 0{,}2)\,\mathrm{S} \\
\underline{Y}(p = 1) &= \left(0{,}05 + \frac{1 + \mathrm{j}}{2} \cdot 0{,}2\right)\,\mathrm{S} = (0{,}15 + \mathrm{j} \cdot 0{,}1)\,\mathrm{S} \\
\underline{Y}(p = 2) &= \left(0{,}05 + \frac{2 + \mathrm{j}}{5} \cdot 0{,}2\right)\,\mathrm{S} = (0{,}13 + \mathrm{j} \cdot 0{,}04)\,\mathrm{S} \\
\underline{Y}(p \to \infty) &= \left(0{,}05 + \frac{\infty + \mathrm{j}}{\infty^2 + 1} \cdot 0{,}2\right)\,\mathrm{S} = (0{,}05 + \mathrm{j} \cdot 0)\,\mathrm{S}
\end{aligned}$$

b) Sketch of the admittance locus curve:



c) Current is 45° ahead for $\varphi = 45°$. Intersection of straight line and locus curve.

d) With $\arctan(45°) = \frac{\mathrm{Im}\{\underline{Y}\}}{\mathrm{Re}\{\underline{Y}\}} = 1$ comes:

$$\begin{aligned}
\mathrm{Re}\{\underline{Y}\} &\overset{!}{=} \mathrm{Im}\{\underline{Y}\} \\
0{,}05 + 0{,}2 \cdot \frac{p}{p^2+1} &\overset{!}{=} 0{,}2 \cdot \frac{1}{p^2+1} \\
0{,}05 \cdot p^2 + 0{,}2 \cdot p - 0{,}15 &= 0 \\
p_{1/2} &= -2 \pm \sqrt{4+3} \\
\text{mit } R_1 > 0: \quad p_{45°} &= -2 + \sqrt{7} \\
\underline{Y}(\varphi = 45°) &= 0{,}05 + 0{,}2 \cdot \frac{-2+\sqrt{7}+\mathrm{j}}{(-2+\sqrt{7})^2+1} \\
&= (0{,}14114 + \mathrm{j} \cdot 0{,}14114)\,\mathrm{S}
\end{aligned}$$

B.9 AC measuring bridge

a) Measured voltage from potential difference of voltage dividers:

$$\begin{aligned}
\underline{U}_B &= U_0 \cdot \left(\frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} - \frac{\underline{Z}_4}{\underline{Z}_3 + \underline{Z}_4} \right) \\
&= U_0 \cdot \left(\frac{\underline{Z}_2 \cdot \underline{Z}_3 + \cancel{\underline{Z}_2 \cdot \underline{Z}_4} - \underline{Z}_1 \cdot \underline{Z}_4 - \cancel{\underline{Z}_2 \cdot \underline{Z}_4}}{(\underline{Z}_1 + \underline{Z}_2) \cdot (\underline{Z}_3 + \underline{Z}_4)} \right) \\
&= U_0 \cdot \left(\frac{\underline{Z}_2 \cdot \underline{Z}_3 \; - \; \underline{Z}_1 \cdot \underline{Z}_4}{(\underline{Z}_1 + \underline{Z}_2) \cdot (\underline{Z}_3 + \underline{Z}_4)} \right)
\end{aligned}$$

b) As in a) only with $\underline{Z}_1 = \mathrm{j}X_1$, $\underline{Z}_2 = \mathrm{j}X_2$, $\underline{Z}_3 = \underline{Z}_4 = R$.

$$\begin{aligned}
\underline{U}_B &= U_0 \cdot \left(\frac{X_2 - X_1}{(X_1 + X_2) \cdot (1+1)} \right) \\
\underline{U}_B &= U_0 \cdot \left(\frac{\mathrm{j}\cancel{R}}{\cancel{\mathrm{j}}R} \cdot \frac{\mathrm{j}X_2 \cdot R - \mathrm{j}X_1 \cdot R}{(\mathrm{j}X_1 + \mathrm{j}X_2) \cdot (R+R)} \right) \\
&= \frac{U_0}{2} \cdot \frac{X_2 - X_1}{X_2 + X_1}
\end{aligned}$$

c) The measured voltage $\underline{U}_B$ is independent of the frequency, provided that there are two capacitive ($X_C \sim \frac{1}{\omega}$) or two inductive reactances ($X_L \sim \omega$). Then the frequency-dependent terms of the reactances X_1 and X_2 cancel each other out.

d) For $X_1 = X_2$, $\underline{U}_B = 0$. Both voltage dividers (for $\underline{U}_B^+$ and $\underline{U}_B^-$) yield $U_0/2$.

e) For $X_1 = -X_2$, the resonance condition is satisfied (of the ideal LC series resonant circuit consisting of X_1 and X_2). The measured voltage $\underline{U}_B$ approaches infinity (since it is undamped).

Literatur

[1] xkcd. *Webcomic*. URL: https://xkcd.com/ (besucht am 25. 02. 2025).

[2] Wilfried Weißgerber. *Elektrotechnik für Ingenieure 2*. dt. 10. Auflage. Springer Vieweg, 2018. ISBN: 978-3-658-21822-2.

[3] Ehrhard Behrends. *Parkettierungen der Ebene - von Escher über Möbius zu Penrose*. dt. Springer Spektrum, 2019. ISBN: 978-3-658-23269-6.

[4] Hans Hagmann. *Grundlagen der Elektrotechnik*. dt. 18. Auflage. AULA-Verlag, 2020. ISBN: 978-3-89104-830-6.

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